Recent Changes in Firm Dynamics and the Nature of Macroeconomic Trends*

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Abstract

Many advanced economies have experienced similar macroeconomic trends in recent decades, e.g., a rise in industry concentration and a fall in firm entry and productivity growth. The literature proposes various causes for these trends. This paper presents novel firm-level evidence to test these explanations. Using Swedish and U.S. administrative data, I show that firm size relative to entry – the difference between firm size at a given age and entry – has systematically increased around the turn of the millennium. I build a comprehensive model that accommodates many forces emphasized in the literature and estimate its parameters to rationalize the firm-level and macroeconomic trends jointly. The estimation highlights a fall in firm expansion costs and a decline in the quality of innovations since the 1990s. Various other forces do not significantly affect firm size relative to entry or imply counterfactual changes in the macroeconomy when explaining the firm-level trends quantitatively.

Keywords: Firm dynamics, Aggregate productivity growth, Administrative data

JEL codes: D22, E24, O31, O47, O50

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1 Introduction

Many advanced economies have experienced similar macroeconomic trends over the past decades, e.g., a rise in industry concentration and a fall in firm entry and aggregate productivity growth.[1] The literature highlights numerous explanations for these trends, e.g., increasing firm entry costs, rising dispersion of firm productivity, or falling costs of firm expansion. Despite the differences in their nature, these forces generally operate through similar mechanisms in models of creative destruction, generating either an expansion of incumbent firms into new product markets or an expansion relative to their competitors within product markets.[2] Yet, there is little empirical evidence documenting such firm-growth dynamics. One important contribution is Hsieh and Rossi-Hansberg (2023), showing that U.S. services firms increasingly operate in small and mid-sized cities. Complementary to this geographic angle of firm size, I document the expansion of firms using general measures of firm size (e.g., firms' value added or the wage bill) in this paper. With the help of a quantitative model that accommodates many of the highlighted causes behind the macroeconomic trends, I use the documented size expansion of firms as a new identifying moment to test the different explanations.

The first contribution of this paper is empirical. Using high-quality administrative data on the universe of firms in Sweden from 1997–2017, the main setting of this paper, I document key trends in the economy. At the macroeconomic level, the rate of aggregate productivity growth and firm entry have fallen by 0.4 and 4 percentage points, respectively, since the late 1990s, which is in line with recent trends in many advanced economies. At the same time, industry concentration, as measured by the interquartile range of value added, has more than doubled. The changes in the macroeconomy are most pronounced during the late 1990s and early 2000s. Using the same data, I document a novel trend at the firm level: firm size relative to entry, i.e., the difference between firm size at a given age and entry, has systematically increased over time. For example, whereas (real) value added at age eight was roughly 49 percent larger than at age zero for firms established in the 1990s, the relative size increased to about 67 percent for the cohorts of the 2010s. For the firms' wage bill, these differences are similar. The relative wage bill at age eight increased from about 47 percent for firms established in the 1990s to roughly 63 percent for the cohorts of the 2010s. These trends are most pronounced for the firms established during the late 1990s and early 2000s, indicating that the trends at the firm and aggregate levels are directly related.

[1] For the U.S., these trends are documented in (among many others) Akcigit and Ates (2021); Andrews, Criscuolo and Gal (2015); Autor, Dorn, Katz, Patterson and Van Reenen (2020); Decker, Haltiwanger, Jarmin and Miranda (2020); Grullon, Larkin and Michaely (2019); Van Reenen (2018); Decker, Haltiwanger, Jarmin and Miranda (2016); Gourio, Messer and Siemer (2014). Andrews, Criscuolo and Gal (2016); Gopinath, Kalemli-Özcan, Karabarbounis and Villegas-Sanchez (2017); Autor, Dorn, Katz, Patterson and Van Reenen (2020); Karabarbounis and Neiman (2014) document macroeconomic trends for other advanced economies.

[2] Examples of incumbent firms expanding into new product markets include Aghion, Bergeaud, Boppart, Klenow and Li (2023); De Ridder (2024). Examples of incumbent firms expanding relative to their competitors within product markets include Liu, Mian and Sufi (2022); Olmstead-Rumsey (2019); Akcigit and Ates (2023).

Motivated by the empirical findings, I build a quantitative model to explain the increase in firm size relative to entry and the changes in the macroeconomy jointly. At its core, the model is an endogenous growth framework of creative destruction in the spirit of the quality-ladder models in Aghion and Howitt (1992) and Klette and Kortum (2004).[3] Incumbent firms and potential entrants gain market shares by improving the quality of their competitors in new product markets through innovations (expansion R&D). The model further includes several features highlighted in the literature as the cause behind the recent macroeconomic trends. In particular, in addition to the expansion into new product markets, incumbents can further increase the quality of their own products through innovations (internal R&D), thereby distancing their competitors vertically within product markets. Moreover, incumbent firms differ exogenously in their production efficiency. Some firms are systematically more efficient at producing than others. This comprehensive model allows me to test different explanations for the increase in firm size relative to entry, e.g., a fall in the firm expansion costs or an increase in the productivity dispersion (as highlighted by Aghion, Bergeaud, Boppart, Klenow and Li, 2023), rising entry costs (as emphasized by Davis, 2017; Gutiérrez and Philippon, 2018), a fall in the step size of quality improvements, i.e., innovations becoming more incremental (Olmstead-Rumsey, 2019), or falling costs of distancing competitors within product markets (akin to the decline in knowledge diffusion in Akcigit and Ates, 2023).

I estimate the model on a balanced growth path, targeting firm size relative to entry (value added and the wage bill) of the cohorts of the late 1990s, as well as aggregate productivity growth, firm entry, and industry concentration at that time. Using simple comparative statics in the estimated model, I first show that falling expansion costs, rising productivity dispersion or entry costs, innovations becoming more incremental, or falling costs of distancing competitors within product markets are each qualitatively consistent with an increase in firm size relative to entry. Hence, common forces highlighted in the literature as drivers behind the aggregate trends are all consistent with an increase in firm size relative to entry.

To investigate which of these stories explains the increase in firm size relative to entry quantitatively, I estimate the model parameters jointly on a new balanced growth path, targeting the increase in firm size relative to entry of the most recent cohorts as well as the rise in industry concentration and the fall in aggregate productivity growth and firm entry observed in the data. Comparing parameter values across balanced growth paths shows that the estimation asks for a fall in the firm expansion costs, a rise in the productivity dispersion, a decrease in the cost of distancing competitors within product markets, and a fall in the step size of quality improvements. In contrast to the simple comparative statics, the joint estimation does not ask for an increase in the entry costs.

To quantify the contribution of each parameter to the changes in the targeted moments, I run a horse race, implementing each estimated parameter change in isolation. The decomposition highlights two forces in particular behind the increase in firm size relative to entry: the decrease in the expansion costs and the fall in the step size of quality improvements. At the

[3] These models capture firm dynamics well (Lentz and Mortensen, 2008; Akcigit and Kerr, 2018).

same time, these two forces explain most of the observed increase in industry concentration and the fall in aggregate productivity growth and firm entry. The estimation identifies these two forces as the main drivers because firm size relative to entry is sensitive to both. A rise in productivity dispersion or a fall in the cost of distancing competitors within product markets has a limited effect. A rise in the entry costs, instead, could rationalize the increase in firm size relative to entry quantitatively; however, the implied decrease in the firm entry rate far exceeds the four percentage point decline in the data. Hence, the increase in firm size relative to entry identifies a fall in the firm expansion costs and a decrease in the step size of quality improvements as the main drivers behind the highlighted changes in the macroeconomy between the 1990s and today.

Lastly, despite the aggregate long-run growth rate falling across the estimated balanced growth paths, the implications for the short run are not obvious, as some of the parameter changes increase the long-run growth rate in isolation (e.g., the fall in the expansion costs). I solve for the full transition path toward the new balanced growth path to study the short-run dynamics of the economy and quantify the implications for welfare. Indeed, following the introduction of the parameter changes as shocks starting from the initial balanced growth path, aggregate consumption booms for about six years, after which it falls below the initial balanced growth path trend. The fall in long-run growth dominates the short-run boost, resulting in a welfare loss of 4.95 percent in permanent consumption units.

I provide further extensions to the empirical and model results. On the empirical side, I extend the findings to the U.S. economy. I use the Business Dynamics Statistics (BDS) provided by the U.S. Census Bureau to show that firm employment relative to entry has systematically increased in the 1990s. One advantage of the Swedish data is that it captures information on firms' value added and the wage bill. However, I show that the increase in employment relative to entry in the U.S. is quantitatively comparable to the documented trends in Sweden. On the model side, I further decompose the increase in firm size relative to entry into changes in firm growth (conditional on survival) and selection among surviving firms. The model highlights that the rate of expansion into new product markets has increased for more productive firms while it decreased for less productive ones. Both increase size growth conditional on survival of more productive firms and their share among surviving firms at any age, rationalizing the increase in firm size relative to entry.

Related Literature. This paper relates to the literature that studies the cause behind recent macroeconomic trends. I contribute to this literature using a novel observation, namely the increase in firm size relative to entry, to test the different theories cited previously. Two further explanations in the literature that are not directly modeled in the framework in this paper highlight the increasing importance of intangible capital in firms' production (De Ridder, 2024; Chiavari and Goraya, 2020; Weiss, 2019) and the fall in population growth (Hopenhayn, Neira and Singhania, 2022; Karahan, Pugsley and Şahin, 2024; Peters and Walsh, 2021). First, I find that, at least for Sweden, the increase in firm size relative to entry is not associated with an increase in the capital stock relative to entry, even when including intangible assets in the capital measure. Arguably, intangible assets in balance sheet data are an imperfect measure of intangible capital, but I take this evidence as motivation not to include intangible capital in the model. Second, the trends at the firm and aggregate level that I explain in this paper occur mostly over a period of ten years starting from the mid-1990s, a time when the IT revolution introduced drastic changes to the economy. I view changes in population growth as an unlikely driver behind the aggregate trends during this relatively narrow time period. Therefore, the causes identified in this paper are complementary to the fall in population growth that transformed the economy even before the mid-1990s.

Other studies have analyzed firm size dynamics over time. Sterk, Sedláček and Pugsley (2021) document changes in firm growth in the U.S. over time. For the firms established in the 1980s, the authors show that employment growth over the firm's life cycle slowed. The results presented in this paper are complementary rather than contradictory to theirs as I focus on firms established from the mid-1990s (and macroeconomic trends) onward. Hopenhayn, Neira and Singhania (2022) and Karahan, Pugsley and Şahin (2024) document that firm employment conditional on firm age has been relatively stable or somewhat declining in U.S. Census data. Rather than looking at levels, I study firm size conditional on age *relative to entry* (age zero) in this paper. In certain cases, employment levels can be misleading. For example, employment levels are more heavily affected by aggregate shocks outside the focus of this literature (the China Shock or the Great Recession) than employment relative to entry. Firm size relative to entry differences out shocks that affect firms of all ages equally. At the same time, firm size relative to entry is a measure of firm growth. In models of endogenous growth, as in this paper, firm growth maps into aggregate growth such that trends in firm size relative to entry are informative about trends in aggregate growth.

The structural model in this paper is, on purpose, kept comparable to others in the literature to ensure a fair comparison between the highlighted mechanisms. The model relates most closely to Aghion, Bergeaud, Boppart, Klenow and Li (2023) and Peters (2020). Aghion, Bergeaud, Boppart, Klenow and Li (2023) build a model of creative destruction with innate (ex-ante) heterogeneity in firm productivity. Their model abstracts from internal R&D and firm entry and exit. Firm entry and exit introduce the life cycle dimension to firms and allow one to study trends in firm size relative to entry. Peters (2020) builds an endogenous growth model in which firms conduct expansion and internal R&D. His model abstracts from ex-ante heterogeneity of firms. In the model in this paper, the size growth of ex-ante, more productive firms accelerates in response to the estimated parameter changes, whereas it slows for less productive ones, highlighting the differentiation of firm types when studying changes in firm growth and survival. I further contribute to the theoretical insights in Peters (2020) by solving for the full transition path toward a new balanced growth path computationally. The computation of the transition path allows for welfare comparisons.[4]

[4]Another technical difference to Peters (2020) is that heterogeneity in productivity introduces a new state variable to the firm's value function, namely the distribution of firm productivity types in the economy. Firms keep track of this distribution to build markup expectations for their decision to enter new product lines. Hence, ex-ante heterogeneity in productivity across firms conceptually differs from ex-ante heterogeneity in, e.g., R&D costs.

The paper proceeds as follows. Section 2 documents the changes in the Swedish economy, including the trends in firm-size dynamics. Section 3 lays out the model that I estimate on separate balanced growth paths in Section 4 to study the causes behind the documented trends. The transitional dynamics are computed in Section 5. Section 6 provides robustness, and Section 7 concludes.

2 Trends in the Swedish economy

This section documents salient trends in the Swedish macroeconomy and the dynamics of firm size. I describe the data first.

2.1 Data

Data is provided by Statistics Sweden, the official statistical agency in Sweden. The main data set is *Företagens Ekonomi*, which covers information from balance sheets and profit and loss statements for the universe of Swedish firms at an annual frequency covering the period 1997-2017. If not mentioned otherwise, I focus on the unbalanced panel of firms. The data contains two measures of firm size that I use throughout the paper: value added and the wage bill. For a detailed description of the variables, see Section A in the Appendix. The data further contains information on the firm's industry and legal type. I restrict the analysis to firms operating in the private economy. The firm's birth year is defined as the year the first worker is associated with a firm (employed or self-employed). I obtain this information from the auxiliary data set *Registerbaserad Arbetsmarknadsstatistik*, containing the universe of employer-employee matches. Firms entering late during the year compare small, on average, to firms entering early. To prevent the timing of entry from affecting the average size of entrants within a year, I define the latter as firms aged zero or one. I further restrict the analysis to firms that eventually hire at least one employee, which filters out never-employing firms that solely exist for tax purposes.

Starting in 2004, the data includes unincorporated self-employed with negative profits. These firms were previously not recorded, leading to an inflow of new firms in that year. The change in the firm entry rate in 2004 is hardly distinguishable from the trend in the cleaned data, which implies that the stock of these firms is very small. Nevertheless, since I cannot differentiate between firms established in 2004 and firms entering the data in 2004 with an earlier birth year, I drop the cohort of firms entering the data in 2004. For the results of the paper, the change in the data is inconsequential, as the trends I document mostly occur before 2004. Further, the characteristics of entrants have been very stable over time, as shown below, indicating no systematic changes in the type of firms entering the data.

Table 1 reports distributional statistics of firm sales, value added, and production inputs in the data (1997 to 2017). Throughout the paper, nominal variables are deflated to 2017 Swedish Krona (SEK) using the GDP deflator. The median firm lists sales of roughly 1.4 million SEK (approx. 0.14 million US dollars) and employs one full-time worker. The

Table 1: Summary statistics (1997-2017)

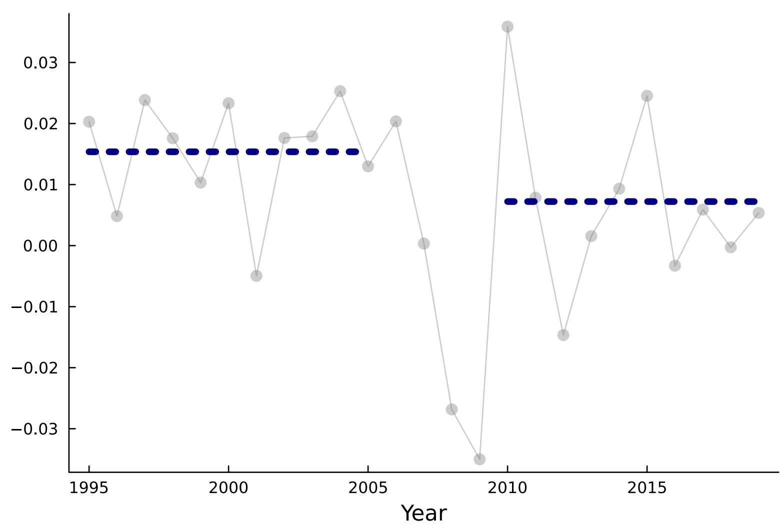
	25th Pct.	50th Pct.	75th Pct.	Mean	SD	Obs.
*Sales**	0.5	1.4	4.4	18.7	464.8	7,374,865
*Value added**	0.2	0.6	1.7	5.1	116.4	7,374,865
Employment (full-time units)	0	1	3	6.6	107.3	7,374,865
*Wage bill**	0.002	0.24	0.87	2.45	43.4	7,374,865
*Capital stock**	0.02	0.16	0.8	6.9	228.5	7,374,865
*Intermediate Inputs**	0.2	0.5	1.6	7.3	220.8	7,374,865

Note: variables marked with * are in units of million 2017-SEK (1 SEK ≈ 0.1 US dollars). The capital stock is defined as fixed assets minus depreciation.

distribution of sales, value added, and all production inputs is highly right-skewed. Mean sales (18.7 million SEK) and full-time employment (6.6) far exceed their respective 75th percentiles. In total, the data includes about 7.4 million firm-year observations. For the firm age- and size-specific empirical analysis, I filter value added at its 0.5% tails and focus on firms established in 1997 or later, which reduces the sample size to 3.5 million firm-year observations.[5] For firms established in 1997 and later, age is not truncated.

2.2 Trends in the macroeconomy

Figure 1: Total Factor Productivity growth



Notes: the scatter points in gray show annual Total Factor Productivity (TFP) growth. The dashed blue line shows the respective averages over 1995–2005 and 2010–2019. Data source: Federal Reserve Economic Data (FRED).

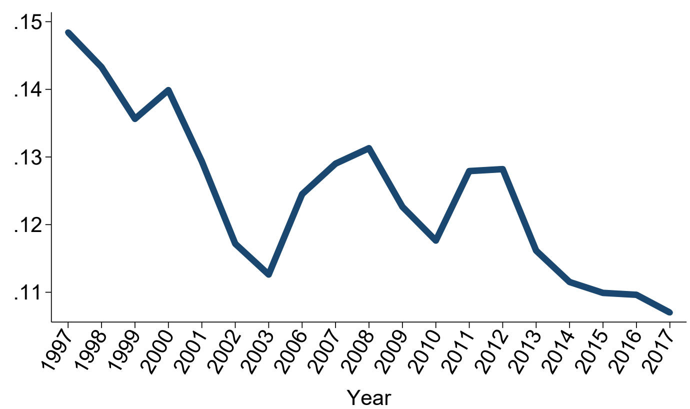
Many economies worldwide have experienced similar macroeconomic trends over the last decades. The Swedish economy is no exception. This section documents trends in aggregate

[5] I apply the 0.5% filter also to the wage bill and the capital stock, but only at the right tail. This ensures that self-employed without labor costs or firms without any physical capital are included in the analysis.

productivity growth, firm entry, and industry concentration in Sweden. For all empirical results in the following section, I leverage the Swedish microdata with one exception. Aggregate productivity growth is obtained from Federal Reserve Economic Data (FRED).

To start with, Figure 1 shows annual Total Factor Productivity (TFP) growth in Sweden from 1995–2019. The data is shown as scatter points, and the dashed blue line refers to time averages. Sweden's economy was characterized by high TFP growth from the mid-1990s to the mid-2000s. Over the period 1995–2005, TFP growth averaged 1.5%. After the financial crisis, TFP growth slowed, averaging about 1.1% over the years 2010–2015. A high growth period during the turn of the millennium, followed by a slowdown after the Great Recession is common across many industrialized countries, as shown, e.g., for the U.S. in Aghion, Bergeaud, Boppart, Klenow and Li (2023).

Figure 2: Entry rate

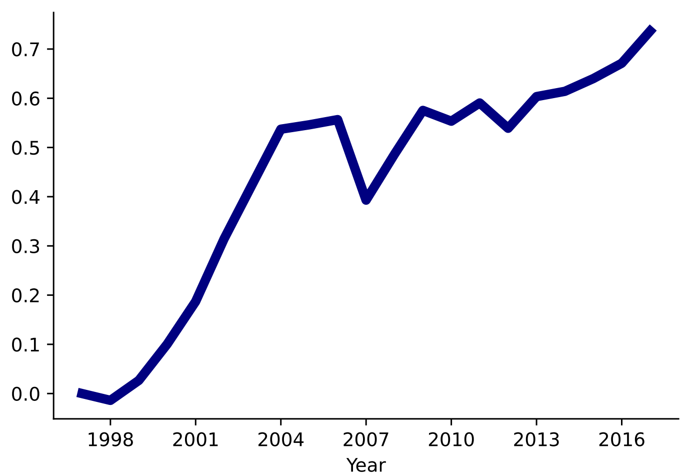


Notes: the graph shows the entry rate defined as the total number of entrants divided by the total number of firms. The years 2004 and 2005 are not shown as I exclude the cohort of firms entering in 2004 and entrants are defined as firms aged zero or one.

Figure 2 shows the evolution of the firm entry rate from 1997–2017, defined as the total number of entrants relative to the total number of firms. The firm entry rate shows a decline from about 15% in 1997 to 11% in 2017. Most of this four-percentage point decline occurs around the turn of the millennium and before the year 2004. After the Great Recession, the entry rate slightly rebounded but declined again shortly after to its previous levels and further below. The years 2004 and 2005 are not shown in the graph, as I exclude the cohort of firms entering in the year 2004, as described in the previous section, and entrants are defined as firms aged zero and one. To be explicit, all year labels are displayed.

Lastly, Figure 3 shows the evolution of industry concentration. As the measure of industry concentration, I use the interquartile range, defined as the difference between the 75th and the 25th percentile, of log value added within two-digit industries. The interquartile range has the advantage over other common measures of industry concentration (e.g., industry shares) or dispersion (e.g., standard deviation) in that it accommodates negative values. At the two-digit industry level, the 25th percentile of the value added distribution is commonly positive such that negative values are not a limiting factor for the interquartile range. At the same time, it is a very robust measure unaffected by outliers. I use the total labor costs to aggregate the interquartile range across industries and normalize the time series to zero in 1997. Figure 3 shows a drastic increase in the size dispersion of firms within industries. Over the period 1997–2017, value added of the 75th percentile firm increased by 0.737 log points relative to the 25th percentile firm. In other words, if the 75th and 25th percentile were identical in 1997, the former would be $e^{0.737} \approx 2.09$ times the size of the latter in 2017. The size dispersion increases particularly strongly during the turn of the millennium: 0.54 log points of the total increase occur before 2004.

Figure 3: Industry concentration



Notes: industry concentration is measured using the interquartile range of log value added within two-digit industries. The measure is aggregated across industries using total labor costs.

2.3 Trends in the dynamics of firm size

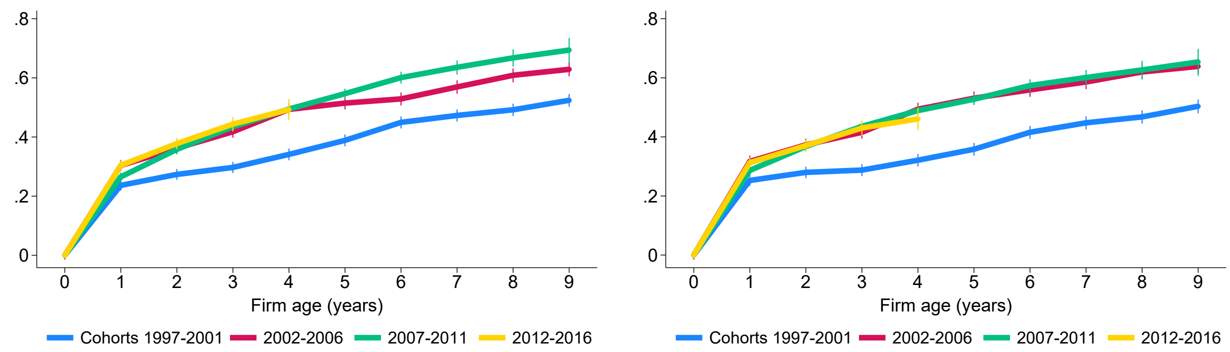
This section documents systematic changes in the dynamics of firm size. To start with, I characterize firm size as a function of firm age nonparametrically. Figure 4 shows the average firm size conditional on survival as a function of firm age. I normalize by the average firm size at entry and show the relative size in logs. The left panel uses value added as the measure of firm size, and the right panel uses the firm's wage bill. The figure includes 95% confidence bands around the means.

As expected, firm size increases in firm age, i.e., older firms are, on average, larger than younger firms. Most importantly, however, the relationship between firm age and size has steepened over time. Figure 4 shows the firm size trajectories for the cohorts 1997–2001, 2002–2006, 2007–2011, and 2012–2016 separately. Compared to the first cohorts 1997–2001, firm size relative to entry has increased for all following cohorts. For example, while value added increased over the first eight years by 0.492 log points for the cohorts of 1997-2001,

Figure 4: Firm size relative to entry (by cohort)

(a) Value added

(b) Wage bill



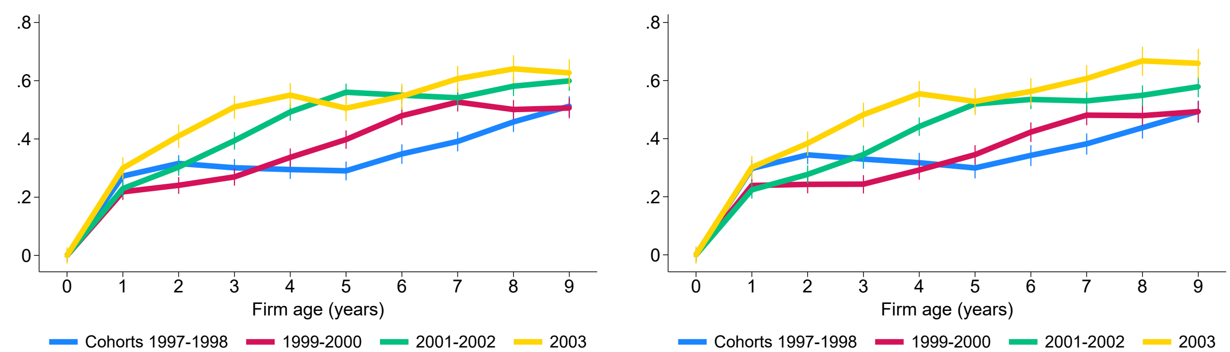
Notes: the figure shows the average value added (left) and wage bill (right) of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. The cohort 2004 is excluded as explained in the main text. 95% confidence bands are included.

the cohorts 2007-2011 experienced an increase of 0.668 log points. The acceleration of size growth is similar for the firm's wage bill (0.468 compared to 0.627 log points for the cohorts 1997–2001 and 2007–2011, respectively).

Figure 5: Firm size relative to entry (cohorts 1997–2003)

(a) Value added

(b) Wage bill

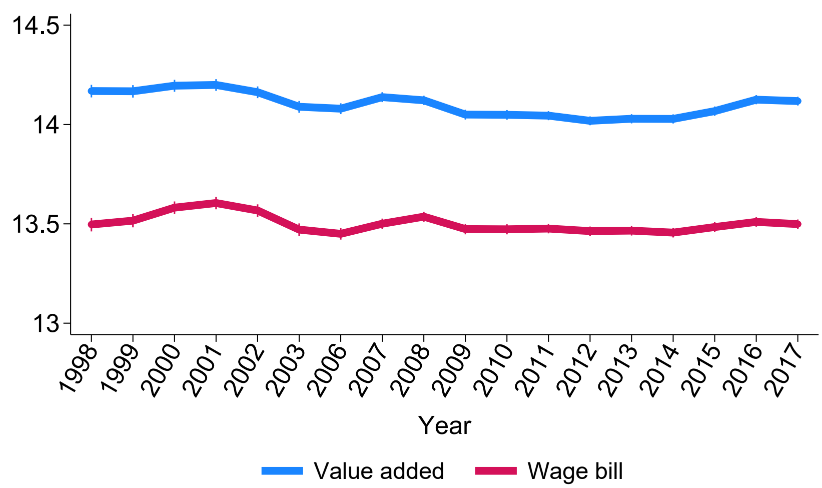


Notes: the figure shows the average value added (left) and wage bill (right) of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. 95% confidence bands are included.

Figure 4 suggests that the changes in firm size relative to entry occurred mostly for the cohorts of the late 1990s and early 2000s. To show this in more detail, I zoom in on these cohorts. Figure 5 shows the firm size trajectories for the cohorts 1997–1998, 1999-2000, 2001–2002 and 2003 separately. The cohort trajectories are less smooth than in Figure 4 since the averages are calculated over two rather than five cohorts, but the gradual upward shift is clearly visible for both the value added and the wage bill. In fact, the size trajectory of the 2003 cohort is similar to that of the subsequent cohorts shown in Figure 4. To be clear, Figures 4 and 5 document an increase in firm size relative to entry without taking a stance on the reason. In particular, the figures remains silent on whether firm growth or the selection of surviving firms has changed.

Firm size relative to entry has systematically increased over time. At the same time, the size of entrants has been very stable. Figure 6 shows the log of average value added and the wage bill of entering firms. Firm size at entry is very stable, not only in the long run but even at the shorter frequency. The stability of firm size at entry indicates that the nature of entering firms has not changed over time.[6]

Figure 6: Firm size at entry



Notes: the figure shows the average firm size (value added and the wage bill) at entry in logs. The years 2004 and 2005 are not shown as I exclude the cohort of firms entering in 2004 and entrants are defined as firms aged zero or one. 95% confidence bands are included.

I provide further evidence and robustness in the Appendix. For example, Figure A-1 shows firms' physical capital stock relative to entry for all cohorts. The trajectories of the capital stock overlap for the different cohorts, indicating no systematic differences in firms' capital accumulation over time. This suggests that the increase in firm size relative to entry is not driven by capital deepening. This picture does not change even when including firms' intangible assets in the measure of the capital stock, see Figure A-2. I further show the size trajectories of value added and the wage bill for firms in the goods and services sector separately in Figure A-3. The increase in firm size relative to entry occurs in both sectors but is particularly clear for services firms. This suggests that the increase in relative firm size is not driven by factors outside the Swedish economy, e.g., foreign goods demand. That these trends are particularly pronounced for services firms is consistent with the evidence in Hsieh and Rossi-Hansberg (2023) that U.S. services firms geographically expanded into

[6]The series starts in 1998 as I exclude firms established before 1997 in the size-age analysis, and entrants are defined as firms aged zero or one.

small and mid-sized cities. However, the results show that the increase in firm size relative to entry is not only a phenomenon in the services sector.

Taking stock, together with the trends at the macroeconomic level, the changes in the dynamics of firm size paint a consistent picture. Firm size relative to entry has systematically increased, particularly for the cohorts that entered between 1997 and 2004. At the same time, the aggregate economy underwent drastic changes. The changes in the firm size dynamics around the turn of the millennium appear key to understanding the macroeconomic trends.

3 Model

This section outlines an endogenous growth model with rich firm dynamics that later sections apply to study the cause behind the documented trends in the firm-size dynamics.

3.1 Preferences and aggregate economy

Time is continuous. The economy consists of a representative household that chooses the path of consumption C_t and wealth A_t to maximize lifetime utility

$$U = \int_0^\infty \exp(-\rho t) \ln C_t dt,$$

subject to the budget constraint $\dot{A}_t = r_t A_t + w_t L_t - C_t$. ρ denotes the discount rate, r_t the interest rate and w_t the real wage. The household supplies one unit of labor inelastically, i.e., $L_t = 1$. The optimality condition (Euler equation) for the household problem reads

$$\frac{\dot{C}_t}{C_t} = r_t - \rho.$$

Aggregate output is produced competitively using a Cobb-Douglas technology over a continuum of different products indexed by i (time subscripts suppressed)

$$Y = \exp\left(\int_0^1 \ln\left[q_i y_i\right] di\right),$$

where y_i and q_i denote the quantity and quality of product i. Output is consumed entirely such that $Y = C$. Expenditure minimization leads to the standard demand function

$$y_i = \frac{YP}{p_i}. \tag{1}$$

P is defined as the aggregate price index, which I normalize to 1.

3.2 Production

Firms produce in a product market i with a linear technology

$$y_{if} = \varphi_f l_{if},$$

where y_{if} is the amount of product i produced by firm f, l_{if} is the amount of labor hired, and φ_f denotes the productivity of firm f. The firm's productivity is fixed over time t and markets i, which captures the notion that some firms are persistently more efficient at producing than others, e.g., due to a better business plan. As in Aghion, Bergeaud, Boppart, Klenow and Li (2023), firms are of a high or low productivity type, i.e., $\varphi_f \in \{\varphi^h, \varphi^l\}$ where $\varphi^h/\varphi^l > 1$, which I refer to as high- and low-type firms.

3.3 Static allocation

Taking the joint distribution of product qualities and firm productivity as exogenous in this section, I characterize the static allocations at the product, firm and aggregate levels.

3.3.1 Product level

Firms in product market i compete in prices (Bertrand competition). In equilibrium, only the firm with the highest quality-adjusted productivity $q_{if}\varphi_f$ produces product i (henceforth, incumbent). Under Bertrand competition, the incumbent firm engages in limit pricing and sets its price equal to the quality-adjusted marginal costs of the follower (the firm with the second highest quality-adjusted productivity)

$$p_{if} = \frac{q_{if}}{q_{if'}} \frac{w}{\varphi_{f'}}, \tag{2}$$

where f' indexes the follower in product market i. The price that the incumbent sets is increasing in the quality gap between the incumbent and the follower, as eq. (2) shows. Defining the product markup as the output price over marginal costs, it follows

$$\mu_{if} \equiv \frac{p_{if}}{w/\varphi_f} = \frac{q_{if}}{q_{if'}} \frac{\varphi_f}{\varphi_{f'}}. \tag{3}$$

The incumbent's markup for product i is increasing in the quality and productivity gap. The price setting of the incumbent gives rise to the following profits for product i

$$\pi_{if} = p_{if} y_{if} - w l_{if} = Y \left(1 - \frac{1}{\mu_{if}}\right),$$

with labor demand for product i

$$l_{if} = \frac{Y}{w}\mu_{if}^{-1}. \tag{4}$$

Employment in product line i is decreasing in the markup.

3.3.2 Firm level

Firm employment is the sum of employment across the firm's product lines

$$l_f = \sum_{i \in N_f} l_{if} = \frac{Y}{w}\left(\sum_{i \in N_f} \mu_{if}^{-1}\right),$$

where N_f denotes the set of product lines where firm f is the incumbent producer. Firm employment decreases in the markups within each product line but increases in the number of product lines. Hence, holding product markups constant, firms that produce in more product lines feature higher employment. Vice versa, holding the number of product lines constant, firms with higher product markups employ less labor. As sales are equalized across product lines, firm sales are given by $\left|N_f\right|Y \equiv n_f Y$, where n_f denotes the number of products firm f is producing. Hence, firms that produce in more product lines feature higher sales. Lastly, I define the firm markup μ_f as total firm sales over the wage bill wl_f.

3.3.3 Aggregate level

Integrating employment across firms or products yields the total workforce in production:

$$L_P = \int_f l_f df = \frac{Y}{w}\int_0^1 \mu_{if}^{-1} di. \tag{5}$$

Taking logs and integrating eq. (3), one obtains an expression for the wage

$$w = \exp\left(\int_0^1 \ln q_{if} di\right) \times \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right) \times \exp\left(\int_0^1 \ln \mu_{if}^{-1} di\right). \tag{6}$$

To find an expression for aggregate output, insert eq. (6) into eq. (5) to obtain

$$Y = Q\Phi\mathcal{M}L_P, \tag{7}$$

where

$$Q = \exp\left(\int_0^1 \ln q_{if} di\right), \quad \Phi = \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right), \quad \mathcal{M} = \frac{\exp\left(\int_0^1 \ln \mu_{if}^{-1} di\right)}{\int_0^1 \mu_{if}^{-1} di}.$$

Aggregate output Y depends on geometric averages of quality Q and productivity Φ as well as on misallocation $\mathcal{M}$ and production labor L_P. As in Peters (2020), misallocation arises from markup dispersion ($\mathcal{M}$ is bounded by unity from above). In this model, markup dispersion is due to both quality and productivity heterogeneity. The product of Q, Φ and $\mathcal{M}$ captures aggregate Total Factor Productivity (TFP).

3.4 Dynamic firm problem

Firms compete through innovations (R&D) for product markets. There are two different types of R&D: firms increase the quality of their own products through successful internal R&D, whereas, through expansion R&D, the firm improves the quality of a random product of a competing incumbent. Item quality is improved step-wise such that every innovation (either internal or expansion R&D) increases q_i by a factor of λ. As Aghion, Bergeaud, Boppart, Klenow and Li (2023), I assume that the step size of quality improvements exceeds the productivity differential, $\lambda > \varphi^h/\varphi^l$. This assumption ensures that the firm with the highest quality version in a product line is the incumbent producer.[7] Denoting by λ^{Δ_i} the relative qualities of incumbent and second-best firms within a product line, i.e.,

$$\lambda^{\Delta_i} = \frac{q_{if}}{q_{if'}}$$

and by $[\mu_i]$ the set of markups, where firm f is producing, firm profits can be written as

$$\pi_{ft}(n, [\mu_i]) \equiv \sum_{k=1}^{n} \pi(\mu_k) = \sum_{k=1}^{n} Y_t \left(1 - \frac{1}{\mu_k}\right) = \sum_{k=1}^{n} Y_t \left(1 - \frac{1}{\lambda^{\Delta_k}} \frac{1}{\frac{\varphi_{fk}}{\varphi_{f'k}}}\right),$$

where $\pi(\mu_i)$ denote profits in product line i. Incumbent firms choose the rate of internal R&D, I_i, and the rate of expansion R&D, x_i, for each of their product lines, i. When choosing optimal internal and expansion R&D rates, firms take aggregate output Y_t, the real wage w_t, the share of lines operated by high-productivity firms S_t, the interest rate r_t and the rate of creative destruction τ_t as given. Denoting the time derivative by $\dot{V}_t^h()$, the value function of a high-productivity type firm (indexed by h) satisfies the following HJB equation:

$$\begin{aligned}
&r_t V_t^h(n, [\mu_i], S_t) - \dot{V}_t^h(n, [\mu_i], S_t) = \\
&\quad \sum_{k=1}^{n} \underbrace{\pi(\mu_k)}_{\text{Flow profits}} + \sum_{k=1}^{n} \underbrace{\tau_t \Big[V_t^h\left(n-1, [\mu_i]_{i \neq k}, S_t\right) - V_t^h(n, [\mu_i], S_t) \Big]}_{\text{Creative destruction}} \\
&\quad + \max_{[x_k, I_k]} \left\{ \sum_{k=1}^{n} \underbrace{I_k \Big[V_t^h\left(n, [[\mu_i]_{i \neq k}, \mu_k \cdot \lambda], S_t\right) - V_t^h(n, [\mu_i], S_t) \Big]}_{\text{Internal R\&D}} \right.
\end{aligned}$$

[7]Relaxing this assumption would give room for a race for incumbency between low-productivity entrants facing a high-productivity incumbent from which I abstract.

$$+\sum_{k=1}^{n}\underbrace{x_k\Big[S_t V_t^h\left(n+1,[[\mu_i],\lambda],S_t\right)+(1-S_t)V_t^h\left(n+1,\left[[\mu_i],\lambda\cdot\frac{\varphi^h}{\varphi^l}\right],S_t\right)-V_t^h(n,[\mu_i],S_t)\Big]}_{\text{Expansion R\&D}}$$

$$\left.-\underbrace{w_t\Gamma\left([x_i,I_i];n,[\mu_i]\right)}_{\text{R\&D costs}}\right\}.$$

The value of a firm consists of flow profits, research costs, and three parts related to internal R&D, expansion R&D, and creative destruction. At the rate of creative destruction τ_t (determined in equilibrium), the firm loses one of its n products, in which case, it remains with $n-1$ products. At the optimally chosen rate I_k, internal R&D turns out successful (third row), and the firm charges a λ times higher markup on its product according to eq. (3). Alternatively, at the optimally chosen rate x_k, expansion R&D is successful (fourth row), and the firm acquires a new product (n increases by one).

Productivity-type heterogeneity introduces novel elements to the value function of the firm. The share of product lines operated by each productivity type S_t is a state variable that firms keep track of to build markup expectations. When taking over a new product line through expansion R&D (fourth row), the probability of replacing a high-type incumbent is S_t, in which case the high-type entrant charges a markup of λ. With probability $1-S_t$, the replaced incumbent is of the low type, and the high-type entrant charges a markup of $\lambda\cdot\varphi^h/\varphi^l$. Firms take S_t as given; however, they affect it through their expansion R&D efforts x_k in equilibrium. The HJB equation for a low-productivity firm follows the same structure and is listed in the Appendix, Section C.1. The term related to expansion R&D (fourth row) is distinct since low-productivity firms build different markup expectations when entering a new product line.

$\Gamma\left([x_i,I_i];n,[\mu_i]\right)$ denote the R&D costs. For their R&D activities, firms pay a cost of

$$\Gamma\left([x_i,I_i];n,[\mu_i]\right)=\sum_{k=1}^{n}c\left(x_k,I_k;\mu_k\right)=\sum_{k=1}^{n}\left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^\zeta+\frac{1}{\psi_x}(x_k)^\zeta\right]$$

in terms of labor. $\zeta>1$ ensures convexity of the cost function. R&D costs are additively separable to render a closed-form solution of the value function along the balanced growth path. ψ_I and ψ_x scale internal and external R&D costs and capture the R&D efficiency.[8]

Firm entry is determined as follows. Potential entrants produce a flow rate of entry z_t using a technology that is linear in labor: $z_t=\psi_z L_{Et}$, where ψ_z denotes the entry efficiency and L_{Et} research labor of entrants. Entrants improve the quality of a randomly selected product line. The productivity type is realized after entry and assigned with the exogenous probabilities p^h and $1-p^h$, respectively. Entrants start with a one-step quality gap. When $z_t>0$, the free

[8]The incentives for internal R&D decrease with the quality gap as product profits are concave in the markup. I scale the internal R&D costs by the inverse markup to keep internal R&D incentives constant as in Peters (2020). Hence, internal R&D costs decline with successful past internal R&D.

entry condition requires that the expected value of firm entry equals the entry costs

$$p^h E[V_t^h(1, \mu_i)] + (1 - p^h) E[V_t^l(1, \mu_i)] = \frac{1}{\psi_z} w_t, \tag{8}$$

where the expected value of entering as a high- or low-type firm is

$$\begin{aligned} E[V_t^h(1, \mu_i)] =& S_t V_t^h(1, \lambda) + (1 - S_t) V_t^h(1, \lambda \times \varphi^h / \varphi^l) \\ E[V_t^l(1, \mu_i)] =& S_t V_t^l(1, \lambda \times \varphi^l / \varphi^h) + (1 - S_t) V_t^l(1, \lambda). \end{aligned}$$

Labor market clearing requires that production labor L_{Pt} and total research labor L_{Rt} add up to one, the aggregate labor endowment

$$1 = L_{Pt} + L_{Rt} = \frac{Y_t}{w_t} \int_0^1 \mu_{it}^{-1} di + \int_0^1 \left(\mu_{it}^{-1} \frac{I_{it}^\zeta}{\psi_I} + \frac{x_{it}^\zeta}{\psi_x} \right) di + \frac{z_t}{\psi_z}. \tag{9}$$

3.5 Cross-sectional distribution of quality and productivity gaps

The joint distribution of quality and productivity gaps between incumbent and laggard firms across product lines determines economic aggregates in the model. On the one hand, quality and productivity gaps characterize the markup distribution that summarizes labor demand. On the other hand, the joint distribution characterizes the share of product lines operated by each productivity type, which is a state variable in the firm's optimization problem. This section derives the joint distribution of quality and productivity gaps as a function of firm policies, which allows the equilibrium distribution to be solved jointly with the policies.

The distribution of quality and productivity gaps ν is characterized by a set of infinitely many differential equations. For simplicity, I characterize the differential equations for firm-type specific expansion R&D rates, x_t^h and x_t^ℓ, and uniform internal R&D rates, I_t, as proven shortly in Proposition 1 for a balanced growth path.[9] For product lines where the incumbent is at least two quality steps ahead of the laggard ($\Delta \geq 2$), the measure ν follows

$$\dot{\nu}_t \left(\Delta, \frac{\varphi_f}{\varphi_{f'}} \right) = I_t \nu_t \left(\Delta - 1, \frac{\varphi_f}{\varphi_{f'}} \right) - \nu_t \left(\Delta, \frac{\varphi_f}{\varphi_{f'}} \right) (I_t + \tau_t). \tag{10}$$

For product lines where the incumbent is one step ahead ($\Delta = 1$), the measure follows

$$\begin{aligned} \dot{\nu}_t \left(1, \frac{\varphi^l}{\varphi^h} \right) &= (1 - S_t) x_t^l S_t + z_t (1 - p^h) S_t - \nu_t \left(1, \frac{\varphi^l}{\varphi^h} \right) (I_t + \tau_t) \\ \dot{\nu}_t \left(1, \frac{\varphi^l}{\varphi^l} \right) &= (1 - S_t) x_t^l (1 - S_t) + z_t (1 - p^h)(1 - S_t) - \nu_t \left(1, \frac{\varphi^l}{\varphi^l} \right) (I_t + \tau_t) \end{aligned}$$

[9] The distribution along the transition path is characterized in the Appendix, Section D.

$$
\begin{aligned}
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^h}\right) &= S_t x_t^h S_t + z_t p^h S_t - \nu_t\left(1, \frac{\varphi^h}{\varphi^h}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^l}\right) &= S_t x_t^h (1 - S_t) + z_t p^h (1 - S_t) - \nu_t\left(1, \frac{\varphi^h}{\varphi^l}\right)(I_t + \tau_t).
\end{aligned} \tag{11}
$$

Changes in the measure $\dot{\nu}$ are due to inflows and outflows. Outflows arise from successful internal R&D (quality gap increases from Δ to $\Delta+1$) and creative destruction (quality gap is reset to unity). Inflows vary with the quality gap. For $\Delta \geq 2$, inflows into state Δ are due to successful internal R&D in product lines with quality gaps of $\Delta-1$. For $\Delta = 1$, inflows result from creative destruction. For example, the measure of products with a low-type incumbent and high-type second best firm $\nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)$ increases due to low-type incumbents and entrants replacing high-type incumbents, captured by $(1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t$ in eq. (11). From the measure ν, one obtains the share of product lines operated by high-type firms

$$
S_t = \sum_{i=1}^{\infty}\left[\nu_t\left(i, \frac{\varphi^h}{\varphi^h}\right) + \nu_t\left(i, \frac{\varphi^h}{\varphi^l}\right)\right]. \tag{12}
$$

3.6 Balanced growth path characterization

I define a balanced growth path of the economy as follows.

Definition 1. *A balanced growth path (BGP) is a set of allocations* $[x_{it}, I_{it}, \ell_{it}, z_t, S_t, y_{it}, C_t]_{it}$ *and prices* $[r_t, w_t, p_{it}]_{it}$ *such that firms choose* $[x_{it}, I_{it}, p_{it}]$ *optimally, the representative household maximizes utility choosing* $[C_t, y_{it}]_{it}$*, the growth rate of aggregate variables is constant, the free-entry condition holds, all markets clear and the distribution of quality and productivity gaps is stationary.*

Along the balanced growth path, the economy can be characterized in closed form.

Proposition 1. *In the above setup, along a balanced growth path:*

1. *The value of a product line for a firm of productivity type* $d \in \{h, l\}$ *is given by*

$$
V_t^d(1, \mu_i, S) = \frac{1}{\rho + \tau}\left[Y_t\left(1 - \frac{1}{\mu_i}\right) + \frac{\zeta - 1}{\psi_x}(x^d)^\zeta w_t + \frac{\zeta - 1}{\psi_I} I^\zeta w_t \mu_i^{-1}\right] \tag{13}
$$

   where $I \equiv I^h = I^l$*. The value of a firm is the sum of its product line values* $V_t^d(n, [\mu_i], S) = \sum_{i=1}^n V_t^d(1, \mu_i, S)$.

2. *More productive firms expand into new product lines faster than less productive ones* $x^h > x^l$.

3. S_{φ^k, φ^p}*, the constant share of product lines where the incumbent firm is of productivity*

type k and the second-best firm of type p is

$$
\begin{aligned}
S_{\varphi^l,\varphi^h} \equiv \sum_{i=1}^{\infty} \nu\left(i, \frac{\varphi^l}{\varphi^h}\right) =& \frac{(1-S)x^l S + z(1-p^h)S}{\tau} \\
S_{\varphi^l,\varphi^l} \equiv \sum_{i=1}^{\infty} \nu\left(i, \frac{\varphi^l}{\varphi^l}\right) =& \frac{(1-S)x^l(1-S) + z(1-p^h)(1-S)}{\tau} \\
S_{\varphi^h,\varphi^h} \equiv \sum_{i=1}^{\infty} \nu\left(i, \frac{\varphi^h}{\varphi^h}\right) =& \frac{Sx^h S + zp^h S}{\tau} \\
S_{\varphi^h,\varphi^l} \equiv \sum_{i=1}^{\infty} \nu\left(i, \frac{\varphi^h}{\varphi^l}\right) =& \frac{Sx^h(1-S) + zp^h(1-S)}{\tau},
\end{aligned}
$$

which implicitly defines the share of product lines operated by the high-productivity type

$$
S = S_{\varphi^h,\varphi^h} + S_{\varphi^h,\varphi^l} = \frac{Sx^h + zp^h}{\tau}. \tag{14}
$$

4. *The growth rate of aggregate variables is given by*

$$
g = \frac{\dot{Q}_t}{Q_t} = \left(\underbrace{I}_{\textit{Incumbent internal R\&D}} + \underbrace{Sx^h + (1-S)x^l}_{\textit{Incumbent expansion R\&D}} + \underbrace{z}_{\textit{Entry}} \right) \times \ln(\lambda). \tag{15}
$$

Proof. The Appendix, Sections C.1, C.2 and C.3, contains the proofs. □

The value of a product line in eq. (13) consists of three terms: profits for a given markup and the continuation values of expansion and internal R&D. The sum of the three terms is discounted by the discount rate and the rate of creative destruction. The more impatient the household or the higher the risk of replacement, the lower the value of a product line. Importantly, the value of a product line is productivity-type specific. More productive incumbents charge higher markups, which increases the value of a product line. This affects the expansion R&D rates. The optimality condition for expansion R&D (Appendix, eq. A-5) equates the expected value of a product line with the marginal cost of expansion R&D. Hence, in equilibrium, more productive firms pay a higher marginal cost of expansion R&D and expand faster into new product markets, i.e., $x^h > x^l$.

Proposition 1 further shows that S_{φ^k,φ^p}, the share of product lines where the incumbent firm is of type k and the second-best firm of type p, is constant along a balanced growth path. This share equals the fraction of creatively destroyed products at each instant of time, where the new incumbent is of type k and the replaced firm of type p. The share of product lines operated by high-productivity type firms S is equal to the sum of S_{φ^h,φ^h} and S_{φ^h,φ^l}. In particular, eq. (14) can be rearranged to

$$S = \frac{zp^h}{(1-S)(x^l - x^h) + z}, \tag{16}$$

which shows that S depends on the difference in the expansion R&D rates between firm types, $x^l - x^h$. Holding firm entry z fixed, an increase in the expansion rate of high-productivity incumbents must be matched by an equal rise (in absolute terms) in the expansion rate of less-productive firms for S to remain constant. Note the importance of firm entry. With z set to zero, for equation (16) to hold, either x^l equals x^h or there is an exterior solution for S. If $x^l \neq x^h$ and firm entry is zero, the faster expanding, more productive firms eventually take over all product lines. Positive firm entry is necessary for both firm types to co-exist in equilibrium where $x^l \neq x^h$ and S is constant at its interior solution.

The aggregate rate of creative destruction is equal to the sum of firm-type specific expansion R&D rates, weighted by the respective sales shares, and the rate of entry

$$\tau = \int_0^1 x_i di + z = Sx^h + (1-S)x^l + z. \tag{17}$$

Long-run growth results from R&D at the product level. This occurs through successful internal R&D or (incumbents' and entrants') creative destruction. The aggregate arrival rate of innovation is, hence, equal to the sum of the rates of creative destruction τ and internal R&D I. Multiplying the arrival rate by the step size of innovation delivers the aggregate growth rate g, as shown in eq. (15) of Proposition 1. Since expansion R&D rates are heterogeneous, changes in the share of product lines operated by each productivity type, S and $1-S$, affect the aggregate growth rate. Along the balanced growth path, both τ and g are constant.

The stationary distribution of productivity and quality gaps further characterizes the aggregate labor income share, the TFP misallocation measure $\mathcal{M}$, and the aggregate markup. I derive these objects analytically in the Appendix, Section C.2.

To find the balanced growth path, I jointly solve the optimality conditions of the firm (derived in Appendix C.1), the free entry condition, eq. (8), the labor market clearing condition, eq. (9), and the system of differential equations characterizing the distribution of productivity and quality gaps captured in eqs. (10) and (11). Appendix C.4 contains the details.

3.7 Firm dynamics

Firms lose products according to the same stochastic process as in Klette and Kortum (2004). However, in this model, firms add products at systematically different rates as optimally chosen expansion R&D rates vary with the firm's productivity type.[10] The following section

[10]Therefore, the properties related to firm-size growth and survival in Klette and Kortum (2004) hold conditional on the firm type. In particular, conditional on the type, firm size, and growth are unrelated, as

derives productivity-type specific expressions for firm growth and survival probabilities.

3.7.1 Firm sales growth

Firm sales are proportional to the number of products a firm produces. As such, successful expansion R&D increases firm sales. Since optimal expansion R&D rates are productivity-type specific, so is sales growth. Conditional on survival, expected sales growth for a firm with productivity φ_f between age zero and age a_f is

$$E\left[\ln n_f Y|a_f, \varphi_f\right] - E\left[\ln n_f Y|0, \varphi_f\right] = \underbrace{g \times a_f}_{\text{Aggregate growth}} + \underbrace{E\left[\ln n_f|a_f, \varphi_f\right]}_{\text{Firm's product growth}},$$

where n_f is the firm's number of products. The probability of producing n products at age a conditional on survival is $\left(1-\gamma^j(a)\right)\left(\gamma^j(a)\right)^{n-1}$, where $\gamma^j(a) = x^j \frac{1-e^{-(\tau-x^j)a}}{\tau-x^j e^{-(\tau-x^j)a}}$ and $j \in \{h, l\}$. Therefore, expected sales growth conditional on survival equals

$$E\left[\ln n_f Y|a_f, \varphi_f\right] - E\left[\ln n_f Y|0, \varphi_f\right] = \underbrace{g \times a_f}_{\text{Aggregate growth}} + \underbrace{\left(1-\gamma^f(a_f)\right)\sum_{n=1}^{\infty} \ln n \times \left(\gamma^f(a_f)\right)^{n-1}}_{\text{Firm's product growth}}. \tag{18}$$

3.7.2 Firm markup growth

I show in the Appendix, Section C.5 that the expected log markup conditional on firm age a_f for a high-productivity type firm is

$$E\left[\ln \mu_f|a_f, \varphi^h\right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^h|a_f]\right)}_{\text{Quality improvements}} + \underbrace{(1-S) \times \ln\left(\frac{\varphi^h}{\varphi^l}\right)}_{\text{Productivity advantage}}, \tag{19}$$

where $E[a_P^h|a_f]$ is the average product age of a high-type firm conditional on firm age.

The expected firm markup conditional on age consists of two terms. The first term in eq. (19) is akin to Peters (2020) and reflects that internal R&D translates quality improvements at the product level into markup growth at the firm level as it ages. In Peters (2020), this term holds for all firms, whereas in this model, this term is specific to the productivity type of the firm, as the average product age varies by firm type. The second term in eq. (19) captures a new level effect that heterogeneity in productivity introduces. The intuition is

in Lentz and Mortensen (2008). For the unconditional firm-size and growth correlation, two forces are at play. On the one hand, young (small) firms tend to grow quicker due to the survival bias. On the other hand, more productive firms (with faster growth rates) are more likely to end up large. In the estimated model, the majority of the firms are of the high-productivity type. Hence, size is unrelated to growth for most firms.

that if a high-type incumbent faces a low-type second-best firm, it can charge a φ^h/φ^l higher markup, which occurs in expectation in $1-S$ of the incumbent's product lines. Expected markup growth conditional on survival then equals $E\left[\ln\mu_f|a_f,\varphi^h\right]-E\left[\ln\mu_f|0,\varphi^h\right]$.

The expected markup conditional on firm age for a low-productivity type firm follows

$$E\left[\ln\mu_f|a_f,\varphi^l\right]=\underbrace{\ln\lambda\times\left(1+I\times E[a_P^l|a_f]\right)}_{\text{Quality improvements}}+\underbrace{S\times\ln\left(\frac{\varphi^l}{\varphi^h}\right)}_{\text{Productivity disadvantage}}. \tag{20}$$

The first term captures quality improvements through internal R&D. The second term in eq. (20) differs from eq. (19). Low-productivity incumbents face a high-productivity second-best firm in a share S of their product lines. Since $\varphi^l<\varphi^h$, this term is negative.

3.7.3 Firm employment growth

Average employment conditional on age and productivity type is equal to

$$E[\ln l_f|=a_f,\varphi_f]=\ln\left(\frac{Y}{w}\right)+E\left[\ln n_f|a_f,\varphi_f\right]-E\left[\ln\mu_f|a_f,\varphi_f\right].$$

Since $\frac{Y}{w}$ is constant, employment growth conditional on survival is given by

$$E[\ln l_f|a_f,\varphi_f]-E[\ln l_f|0,\varphi_f]=\underbrace{E\left[\ln n_f|a_f,\varphi_f\right]}_{\text{Firm's product growth}}-\underbrace{\left(E\left[\ln\mu_f|a_f,\varphi_f\right]-E\left[\ln\mu_f|0,\varphi_f\right]\right)}_{\text{Firm's markup growth}}, \tag{21}$$

where $E\left[\ln n_f|a_f,\varphi_f\right]$ and $E\left[\ln\mu_f|a_f,\varphi_f\right]-E\left[\ln\mu_f|0,\varphi_f\right]$ are defined in eqs. (18)-(20). As wages and output grow at the same rate along the balanced growth path, eqs. (18) and (21) further imply that growth in the firm's wage bill is equal to sales growth minus markup growth.

3.7.4 Firm survival

Firms that lose their last product exit the economy. Since firm size growth is type-dependent, so is firm survival. The survival function in Klette and Kortum (2004) holds conditional on the firm type, i.e., the share of high and low type firms surviving until age a_f is

$$\chi^h(a_f)=1-\tau\frac{1-e^{-(\tau-x^h)a_f}}{\tau-x^he^{-(\tau-x^h)a_f}}\quad\text{and}\quad\chi^l(a_f)=1-\tau\frac{1-e^{-(\tau-x^l)a_f}}{\tau-x^le^{-(\tau-x^l)a_f}}. \tag{22}$$

Hence, the share of high-type firms among surviving firms at age a_f is

$$s^h(a_f)=\frac{p^h\chi^h(a_f)}{p^h\chi^h(a_f)+(1-p^h)\chi^l(a_f)}, \tag{23}$$

which corresponds to the mass of high-type survivors relative to the total mass of survivors. Since $x^h > x^l$, it is easy to show that sales growth conditional on survival of high-productivity firms exceeds that of low-productivity ones and that the share of the former among surviving firms $s^h(a_f)$ increases in a_f.

3.7.5 Firm size relative to entry

Having defined firm growth and survival probabilities by productivity type allows us to characterize firm size relative to entry. Firm size conditional on age relative to the size at entry equals

$$
\begin{aligned}
E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0\right]= \\
s^h(a_f)\times\underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f,\varphi_f=\varphi^h\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^h\right]\right)}_{\text{Size growth cond. on survival (high-productivity type)}} \\
+\left(1-s^h(a_f)\right)\times\underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f,\varphi_f=\varphi^\ell\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^\ell\right]\right)}_{\text{Size growth cond. on survival (low-productivity type)}} \\
+\left(s^h(a_f)-s^h(0)\right)\times\underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^h\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^\ell\right]\right)}_{\text{Firm-type selection}}.
\end{aligned}
\tag{24}
$$

Firm size conditional on age a_f relative to entry is equal to the sum of firm growth conditional on survival of high- and low-productivity firms over the first a_f years, weighted by their share among surviving firms, $s^h(a_f)$ and $1-s^h(a_f)$ and a firm-type selection term. The last term captures the fact that the share of each productivity type among firms aged a_f is not equal to their share at entry. If firm size is measured using sales, the last term disappears since entrants of both productivity types start with equal sales.

3.7.6 Firm size distribution

The theory makes precise predictions about the firm-size distribution. I derive the firm-size distribution in Section C.6 in the Appendix. Denoting by M^h the mass of high-productivity type firms and by M the total mass of firms, the share of high-productivity type firms is

$$S_{M^h}=\frac{M^h}{M}, \tag{25}$$

and the firm entry rate

$$\text{Firm entry rate}=\frac{z}{M}. \tag{26}$$

4 Balanced growth path application

This section applies the model to explain the documented changes in the firm-size dynamics and the macroeconomy. To this extent, I estimate the model along two balanced growth paths. The initial balanced growth path captures firm size dynamics and aggregate economic conditions during the 1990s. I then re-estimate model parameters to reflect the changes in the firm-size dynamics and aggregate conditions with respect to the 2010s.

4.1 Initial balanced growth path

There are, in total, eight parameters in the model. The internal R&D efficiency ψ_I, the expansion R&D efficiency ψ_x, the innovation cost curvature ζ, the entry efficiency ψ_z, the step size of innovation λ, the productivity differential φ^h/φ^ℓ, the share of high-productivity type firms among entrants p^h, and the discount rate ρ. Three of these parameters are calibrated exogenously. First, the discount rate ρ is set to 0.05. Second, I follow Acemoglu, Akcigit, Alp, Bloom and Kerr (2018) and Peters (2020) that set ζ equal to two based on evidence from the microeconometric innovation literature (Blundell, Griffith and Windmeijer, 2002; Hall and Ziedonis, 2001). Third, Figure 6 shows that firm size at entry has remained very stable over time, indicating no systematic changes in the nature of entrants. Therefore, I fix the share of high-productivity firms among entrants p^h at some exogenous level. Since the interquartile range as the measure of industry concentration in this paper captures dispersion around the median firm, I set p^h to 0.5. Hence, one can interpret high- and low-productivity firms as above or below median firms at entry. Setting p^h exogenously further allows a tight identification of the productivity gap φ^h/φ^ℓ. I test the robustness of the results with respect to the value of p^h.

The remaining five parameters are estimated, targeting moments of firm size as well as cross-sectional firm heterogeneity and economic aggregates. In particular, I target firm size (value added and the wage bill) relative to entry, industry concentration, the firm entry rate, and aggregate TFP growth. Even though all parameters are identified jointly, there is a tight mapping between parameters and targets.

Matching value added and the wage bill relative to entry disciplines the firms' R&D efficiencies ψ_x and ψ_I. In the model, successful expansion R&D translates into firm sales growth. In the estimation, ψ_x adjusts expansion R&D costs such that sales relative to entry match a given target. Since labor is the only production input in the model, firm sales equal value added, and I use the latter as the measure in the data. Moreover, the internal R&D costs govern firms' markup growth. Since firm markup is the wedge between value added (sales) and the wage bill in the model, targeting growth in value added and the wage bill jointly disciplines firm markup growth and, hence, the internal R&D efficiency ψ_I. The advantage of targeting the wage bill instead of markups is that the former is directly observed in the data. I target the average value added and wage bill of firms aged eight relative to entry for the cohorts 1997 to 2001. According to Figure 4, value added increased by 0.492 and the wage bill by 0.468 log points over the first eight years. Eight years is long enough to reflect firm size growth and still allow for estimating separate balanced growth paths (one for the early cohorts and one for the latest cohorts) over the data coverage period from 1997 to 2017. The model matches the firm-size dynamics over the entire life cycle rather well, so the specific age targeted is not consequential.

To pin down the productivity differential φ^h/φ^ℓ, I target the industry concentration observed in the data. To see the link, note that firms' size growth generates cross-sectional size dispersion. To the extent that the observed size dispersion in the data exceeds the model-generated value for the targeted value of firm-size growth, the model asks for an increase in the heterogeneity of size growth across both productivity types by increasing the productivity differential. As the measure of size dispersion, I target the interquartile range of log value added within two-digit industries. Size dispersion in the data is enormous. To make the size dispersion in the data comparable to the model, I normalize a firm's value added by its respective value added at entry. This normalization equalizes firm size at entry and resembles the fact that, as common in Klette and Kortum (2004) style models, all firms start with one product. Without this normalization, all size heterogeneity is loaded on productivity differences, even if size at entry varies for other reasons in the data. Since the year of entry is left-censored in the data, I take the average within-industry interquartile range over the most recent years (2010–2017) and subtract the observed increase in Figure 3. This results in an interquartile range of 0.742, i.e., the 75th-percentile firm is roughly $e^{0.742} \approx 2.1$ times as large as the 25th-percentile firm in 1997. In the model, the interquartile range is restricted to integer values.[11] Since the value in the data is very close to log(2), I target this value for the initial balanced growth path.[12]

Identification of the remaining two parameters is straightforward. The entry efficiency ψ_z determines the cost of a given flow rate of entry in units of labor. I target an entry rate of 14.8%, the value in 1997 in Figure 2, to pin down ψ_z. Lastly, aggregate TFP growth disciplines the step-size of quality improvements λ: the aggregate growth rate g in eq. (15) directly depends on λ. I target an aggregate growth rate of 1.5%, the average over the period 1995–2005 in Figure 1. All targets are summarized in Table 2.

The estimation follows a two-step approach. In the first (global) step, the algorithm computes the sum of squared percentage deviations from the targeted moments for a large Sobol sequence of parameter vectors. All targets receive equal weights. In the second (local) step, I take the best candidates from the first step and perform a local search. The local search, again, minimizes the distance from the targets. The estimation is robust, i.e., the best parameter vectors from the second step converge to the same parameter values.

[11] For example, if the firm at the 25th percentile of the cross-sectional value-added distribution produces one product, the interquartile range of log value added is equal to log(2) if the 75th percentile firm produces two products, log(3) if the 75th percentile firm produces three products and so forth.

[12] To improve the computational efficiency of the estimation, I target a value for the standard deviation of log value added instead of the interquartile range and check afterward that the latter equals log(2) along the balanced growth path. The advantage of the former is that it provides finer gradients of the objective function in the parameter search. Computing the empirical counterpart of the standard deviation would require dropping firms with negative value added.

Table 2: Initial balanced growth path. Moments and parameters

	Data	Model
Moments		
Value added at age 8 relative to entry in logs	0.492	0.492
Wage bill at age 8 relative to entry in logs	0.468	0.468
Interquartile range of log value added	log(2)	log(2)
Entry rate in %	14.8	14.8
TFP growth g in %	1.5	1.5
Parameters		
ψ_x *Expansion R&D efficiency*		0.892
ψ_I *Internal R&D efficiency*		0.925
φ^h/φ^ℓ *Productivity gap*		1.019
ψ_z *Entry R&D efficiency*		3.790
λ *Step size of quality improvements*		1.051
Set exogenously		
ρ *Discount rate*		0.05
ζ *R&D cost curvature*		2
p^h *Share of high-type firms among entrants*		0.5

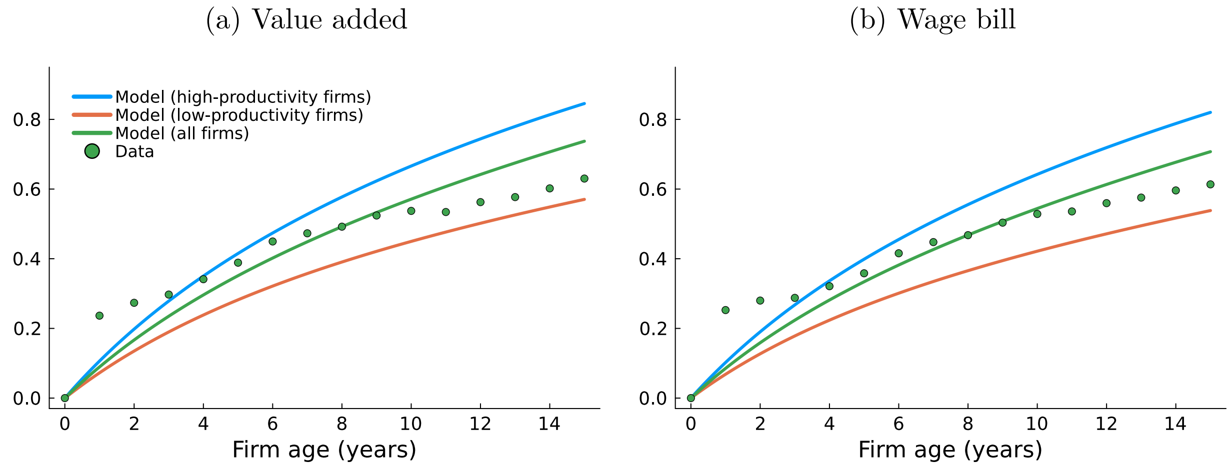
Notes: the targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

Table 2 shows the estimation results. The model replicates all targeted moments exactly. Two parameters can be interpreted intuitively: successful innovation increases product quality by 5.1%, and the productivity differential φ^h/φ^ℓ equals roughly 2%. The productivity advantage is relatively small, which relates to the fact that I target firm-size dispersion in the data that goes beyond dispersion at entry.[13] Nevertheless, firm characteristics differ significantly over the life cycle for the two types, as shown below.

Along the balanced growth path, high-productivity firms constitute 56% of all firms, S_{M^h} in eq. (25), and capture 65% of total value added, as measured by S. These numbers are larger than their share at entry ($p^h = 0.5$) due to high-productivity firms expanding faster into new product lines than less productive ones: $x^h - x^\ell = 0.053$. This is reflected in their size growth. Conditional on survival, value added grows on average by 0.58 log points for high-type firms but only by 0.39 log points for low-type firms over the first eight years. Figure 7 shows the trajectories of value added and the wage bill for each productivity type over the entire life cycle. The figure clearly illustrates the heterogeneity in growth profiles. Conditional on survival, the difference in value added growth over the first 15 years between both productivity types accumulates to roughly 0.3 log points. Figure 7 includes the size trajectories in the data (cohorts 1997–2001). The model slightly misses the concavity of the

[13] Another reason why the productivity gap is relatively small compared to, e.g., Aghion, Bergeaud, Boppart, Klenow and Li (2023) is that their model abstracts from firm markup growth, so the productivity differential plays a larger role in explaining the aggregate markup, which is targeted in their estimation.

Figure 7: Firm size relative to entry



Notes: the figure shows the average value added (left) and wage bill (right) relative to entry (in logs) as a function of firm age in the model (initial balanced growth path) and the data (cohorts 1997–2001 in Swedish registry data). The size at age eight relative to entry has been targeted.

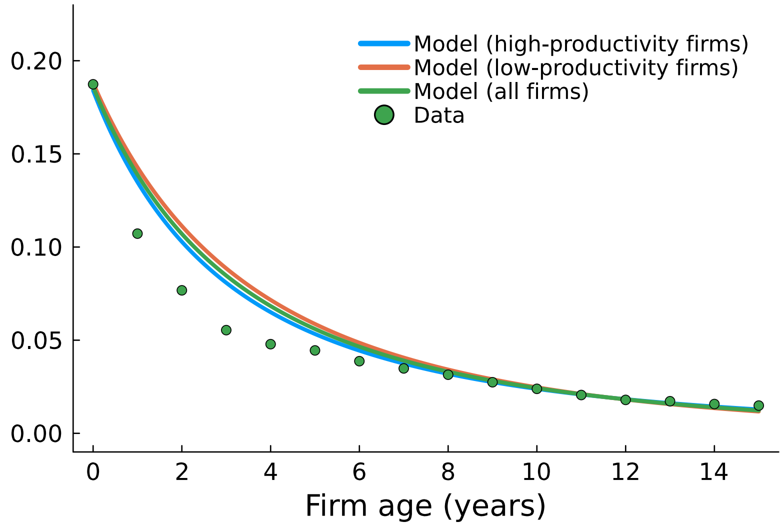
trajectories for the first two years; however, the fit over the entire life cycle is overall good despite being untargeted (except for age eight).

The model also performs well along other untargeted dimensions. Figure 8 shows the firm exit rates by age in the model and the data (cohorts 1997-2001). The model slightly overshoots the exit rates for the first few years, consistent with undershooting firm size growth in Figure 7 for these years. However, the overall fit over the entire life cycle is good. In the model, faster size growth by the more productive firms translates into lower exit rates, as Figure 8 shows. The difference in exit rates between both productivity types accumulates over time. The share of high-productivity firms that survive until age 15 is about five percentage points higher than the share of low-productivity ones.

Next, I perform simple comparative statics starting from the initial balanced growth path to show how changes in the estimated parameters affect firm size relative to entry and macroeconomic outcomes in the model. I change each parameter value separately by 20% and choose the direction of change to generate an increase in firm size (value added) relative to entry (compared to the initial balanced growth path). Table 3 shows selected equilibrium outcomes as well as the targeted moments of the previous estimation. As the interquartile range does not respond to all parameter changes, I show the standard deviation of log value added across firms, denoted by σVA, as the measure of firm size dispersion. The size of firms aged eight relative to entry in logs, targeted in the previous estimation, is shown for value added (denoted VA) and the wage bill (wL).

Table 3 shows that a rise in firm size relative to entry is consistent with many mechanisms highlighted in the literature behind recent macroeconomic trends. In particular, a fall in the expansion costs (increase in the expansion R&D efficiency; row 2), a fall in the cost of distancing competitors within product lines (increase in the internal R&D efficiency; row 3),

Figure 8: Exit rates by firm age



Notes: the figure shows the firm exit rates as a function of age in the estimated model (initial balanced growth path) and the data (cohorts 1997–2001). The exit rates are computed as the increments in a Kaplan-Meier survival function.

an increase in the productivity differential (row 4), a rise in the entry costs (fall in the entry efficiency; row 5) and a fall in the step-size of quality improvements (row 6) cause an increase in firm size (value added) relative to entry, as shown in column 7. Further, the responses of macroeconomic variables are intuitive. For example, the increase in the expansion R&D efficiency and productivity differential and the fall in the entry efficiency and step-size of quality improvements generate rising firm size dispersion (column 9) and falling firm entry (column 10). The effect on the aggregate growth rate (last column) is most pronounced for the fall in the step-size of quality improvements, decreasing g from 1.5% to 1.09%.

Table 3 shows that firm size relative to entry reacts most strongly to the rise in the expansion R&D efficiency and the fall in the entry efficiency. However, the fall in the entry efficiency causes an extreme fall in the firm entry rate (from 14.8% to 7.5%), as further highlighted in the next section. Table 3 further shows that a fall in the step size of quality improvements leads firms to substitute from internal R&D to expansion R&D, at least the more productive ones. At the same time, firm entry falls, lowering the rate of creative destruction τ. As the rate of replacement falls, firm size relative to entry increases. With the exception of the internal R&D efficiency, all shown parameter changes increase both value added and the wage bill relative to entry (compared to the initial balanced growth path). A rise in the internal R&D efficiency increases markup growth, thereby reducing growth in the firm's wage bill relative to value added.

4.2 New balanced growth path

This section estimates the model on a new balanced growth path that replicates the changes in the dynamics of firm size and the macroeconomy vis-a-vis the initial balanced growth path.

Table 3: Comparative statics in the initial balanced growth path

	x^h	x^ℓ	I	τ	S	VA	wL	σVA	$\frac{z}{M}$	g
Initial BGP	0.144	0.091	0.085	0.217	0.63	0.492	0.468	0.5	0.148	0.015
Parameter Δ										
$\psi_x \uparrow$	0.177	0.105	0.083	0.224	0.72	0.581	0.558	0.63	0.134	0.0152
$\psi_I \uparrow$	0.143	0.092	0.1	0.222	0.62	0.494	0.466	0.49	0.154	0.016
$\varphi^h/\varphi^\ell \uparrow$	0.15	0.085	0.085	0.215	0.67	0.499	0.475	0.53	0.144	0.0149
$\psi_z \downarrow$	0.182	0.1	0.095	0.188	0.94	0.624	0.597	1.18	0.075	0.0141
$\lambda \downarrow$	0.154	0.082	0.083	0.179	0.8	0.511	0.49	0.74	0.098	0.0109

Notes: the table shows balanced growth path (BGP) outcomes (columns) for different model parametrizations (rows). x^h and x^ℓ denote the expansion R&D rates of high- and low-productivity firms, I the rate of internal R&D, τ the rate of creative destruction, S the share of product lines operated by high-produtivity firms, VA the difference in average log value added between firms aged eight and entrants, wL the difference in the average log wage bill between firms aged eight and entrants, σVA the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate. The parameters ψ_x, ψ_I and ψ_z denote the expansion, internal and entry R&D efficiency, respectively. φ^h/φ^ℓ is the productivity differential and λ the step-size of quality improvements. The size of the parameter change is 20% (for φ^h/φ^ℓ and λ the change refers to the parameter value minus one). The direction of the change is chosen to generate an increase in ΔVA relative to the initial BGP.

I target changes in the five moments that I used in the estimation of the initial balanced growth path. The first two targets capture the increase in firm size relative to entry (value added and the wage bill) shown in Figure 4. I target the average of the cohorts 2002-2006 and 2007–2011. For value added, this results in a size increase of 0.638 log points over the first eight years and for the wage bill of 0.623 log points. These numbers exceed the respective values in the initial balanced growth path (0.492 and 0.468). As the third moment, I target the change in industry concentration. Figure 3 shows an increase in the interquartile range of value added of 0.74 log points. I target an increase from log(2) in the initial balanced growth path to log(5) in the new balanced growth path. This increase of 0.92 log points slightly exceeds the observed one of 0.74. I target an increase to log(5) rather than log(4) (which would undershoot the observed change) to give the productivity differential φ^h/φ^ℓ a fair chance in the estimation to play a role. Even in this case, it will turn out that changes in the productivity differential play a rather muted role.[14] Lastly, I target the fall in the firm entry rate from 15% to 11% shown in Figure 2 and the decline in the aggregate growth rate from 1.5% to 1.1% shown in Figure 1. I allow all five parameters of the previous estimation to change.

Table 4 captures all targeted moments and parameter estimates of the new balanced growth path. The estimation replicates the changes in all five moments well. The last column of the table compares the estimated parameters with their respective values in the initial balanced growth path. The parameter changes are consistent with the literature that studies the drivers behind recent macroeconomic trends. The estimation highlights a fall in the cost of expanding into new product lines (rise in the expansion R&D efficiency) and an increase

[14] As described for the initial balanced growth path, I target a value for the standard deviation of log value added in the estimation and check afterward that the generated interquartile range equals log(5).

Table 4: New balanced growth path. Moments and parameters

	Data	Model	Initial BGP
Moments			
Value added at age 8 relative to entry in logs	0.638	0.638	0.492
Wage bill at age 8 relative to entry in logs	0.623	0.623	0.468
Interquartile range of log value added	log(5)	log(5)	log(2)
Entry rate in %	11	11	14.8
TFP growth g in %	1.1	1.1	1.5
Parameters			
ψ_x *Expansion R&D efficiency*		1.967	0.892
ψ_I *Internal R&D efficiency*		1.181	0.925
φ^h/φ^ℓ *Productivity gap*		1.021	1.019
ψ_z *Entry R&D efficiency*		6.395	3.790
λ *Step size of quality improvements*		1.036	1.051

Notes: the targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

in the productivity gap between high- and low-type firms. Aghion, Bergeaud, Boppart, Klenow and Li (2023) arrive at a similar conclusion. Akcigit and Ates (2023) argue that a decline in knowledge diffusion allowed incumbent firms to distance competitors within product markets. The increase in the internal R&D efficiency shown in Table 4 highlights a similar mechanism, with the difference that distancing competitors occurs through successful incumbent R&D in this model. Lastly, the estimation points to innovations becoming more incremental as the step size of quality improvements declines. This mechanism is emphasized by Olmstead-Rumsey (2019). Bloom, Jones, Van Reenen and Webb (2020) further show that R&D expenditures have increased relative to R&D output, potentially due to innovations becoming more incremental. The estimated increase in the entry R&D efficiency in Table 4 is somewhat surprising at first, and I will address it shortly.

To quantify how much each parameter contributes to the changes in the five moments, I run a horse race. Starting from the initial balanced growth path, I measure the contribution of parameter $\theta_i \in \boldsymbol{\theta}$ to the change in moment $M_j(\boldsymbol{\theta})$ as

$$\frac{M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{new}\right) - M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{initial}\right)}{M_j\left(\boldsymbol{\theta}_{-i}^{new}, \theta_i^{new}\right) - M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{initial}\right)} \times 100, \tag{27}$$

where *initial* and *new* index the initial and new balanced growth path.[15] Table 5 shows

[15] Setting ψ_x or λ to their values in the new balanced growth path with the entry R&D efficiency ψ_z kept at its initial balanced growth path value results in zero firm entry. Without entry, the firm-size distribution is non-stationary, i.e., no balanced growth path exists. Hence, for ψ_x and λ, I compute the total contribution by multiplying the contribution of a 0.1% change of the total change in the respective parameter by a factor of 1000. In Appendix E, I perform the decomposition starting from the new balanced growth path instead,

the contribution of each parameter change (rows) to the five moments of the estimation (columns). The notation follows Table 3. Naturally, summing the contributions across rows does not return a value of 100% as the model is nonlinear. As for the previous simple comparative statics, I use as the measure of firm size dispersion the standard deviation of log value added across firms, as the changes in ψ_I and φ^h/φ^ℓ are too small to move the interquartile range. The standard deviation provides a finer measure of the change in size dispersion.

Table 5: Decomposition

	VA	wL	σVA	$\frac{z}{M}$	g
Parameter Δ					
Expansion R&D efficiency $\psi_x \uparrow$	348.4	335.5	107.6	164.1	-37.9
Internal R&D efficiency $\psi_I \uparrow$	2.2	-1.5	-3.4	-19.7	-34.0
Productivity gap $\varphi^h/\varphi^\ell \uparrow$	3.2	2.7	2.9	7.1	1.3
Entry R&D efficiency $\psi_z \uparrow$	-127.2	-116.4	-45.5	-355.7	-97.7
Step size of quality improvements $\lambda \downarrow$	7.9	11.6	33.5	181.3	188.2

Notes: the table shows the contribution of each parameter change (rows) to the changes in the five moments (columns) across the intial and new balanced growth path (BGP) in percent. The contribution of each parameter change is computed using eq. (27). Moments and parameter values of the initial and new BGP are shown in Table 4. VA denotes the difference in average log value added between firms aged eight and entrants, wL the difference in the average log wage bill between firms aged eight and entrants, σVA the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate.

Table 5 highlights two parameter changes in particular behind the rise in firm size relative to entry: the increase in the expansion R&D efficiency and the fall in the step size of quality improvements. The increase in the expansion R&D efficiency more than explains the rise in the relative firm size, whereas the fall in the step size of quality improvements accounts for 7.9% (value added, column 2) and 11.6% (wage bill, column 3). That the estimation selects these two parameters to explain the increase in the relative firm size becomes obvious from the simple comparative statics in Table 3: firm size relative to entry increases strongly following an increase in the expansion R&D efficiency or a fall in the step size of quality improvements. An increase in the internal R&D efficiency or the productivity gap has a much more muted effect.

At the same time, the increase in the expansion R&D efficiency and the fall in the step size of quality improvements cause a surge in the cross-sectional firm size dispersion and a fall in firm entry and the aggregate growth rate that more than explain their targeted changes, as shown in Table 5. Hence, matching the observed macroeconomic trends does not require further help from the productivity gap or the internal R&D efficiency. Since the increase in the expansion R&D efficiency and the fall in the step size of quality improvements more than explain the observed fall in firm entry, the estimation even asks for an increase in the entry R&D efficiency.

which allows one to measure the contributions of ψ_x or λ completely nonlinearly.

Given the results of the simple comparative statics in Table 3, one might have expected a fall in the entry R&D efficiency as an outcome of the estimation. In fact, the fall in the entry R&D efficiency in the simple comparative statics is quantitatively consistent with the observed changes in the relative firm size. However, the fall in the entry R&D efficiency in Table 3 lowers the entry rate far below its empirical value. At the same time, the aggregate growth rate does not decline sufficiently. Matching the fall in the aggregate growth rate would, therefore, require a fall in the step size of quality improvements, lowering firm entry even further. As the other parameters cannot raise firm entry enough, the estimation instead selects the expansion R&D efficiency and the step size of quality improvements to explain most of the changes in the moments. The estimated increase in the entry R&D efficiency highlights the importance of the joint identification of the cause compared to the simple comparative statics in Table 3.[16]

As an alternative to the decomposition in eq. (27), one can quantify the contribution of each parameter change starting from the new balanced growth path. I show the results of the alternative decomposition in Appendix E. In the alternative decomposition, the fall in the step size of quality improvements plays a larger role in explaining the increase in firm size relative to entry than in Table 5. However, the role of each parameter change is overall the same.

Taking stock, the increase in firm size relative to entry requires an increase in the R&D efficiency and a fall in the step size of quality improvements that are powerful enough to explain the observed rise in industry concentration and fall in firm entry and productivity growth in the data. Other forces, such as a rising productivity gap, reduced costs of distancing competitors vertically, or increasing entry costs, either do not substantially affect firm size relative to entry or, if they do, imply counterfactually large changes in the macroeconomy.

5 Transitional dynamics

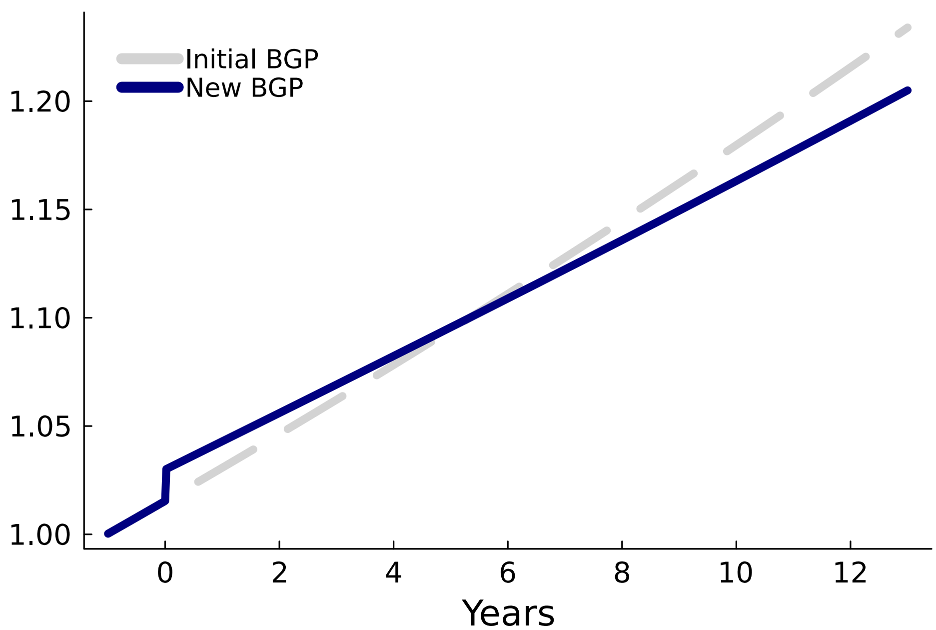
The previous section analyzes the long-run changes in the economy. Even though productivity growth falls across balanced growth paths, some of the estimated parameter changes in isolation increase long-run growth, as the earlier decomposition in Table 5 shows. Which of the parameter changes dominate short-run growth is a quantitative question. Therefore, I solve for the full transition path to the new balanced growth path in this section. The transition path also informs about changes in welfare that are associated with the transition towards the new balanced growth path.

The algorithm to solve for the transition path works as follows. I solve for firms' policy

[16] A common argument for increasing entry costs is that incumbent firms have erected barriers for new firms to enter their markets. One can come up with contrasting arguments, e.g., that the cost of financing has decreased for young firms over time. Which of these forces dominates is a quantitative question and the above model estimation points toward the latter force.

and value functions from the new balanced growth path backward for a guessed sequence of wage growth, interest rates, and market shares of high-productivity firms, S_t. I then use the obtained policy functions over the transition period to simulate the two-dimensional distribution of quality and productivity gaps forward, starting from the initial balanced growth path. Using the evolution of this distribution over the transition, as well as market clearing and optimality conditions, I back out the implied sequences of wage growth, interest rates, and S_t. The transition path is the fixed point between the guessed and implied sequences. I outline the algorithm in detail in the Appendix, Section D.

Figure 9: Transitional dynamics for output



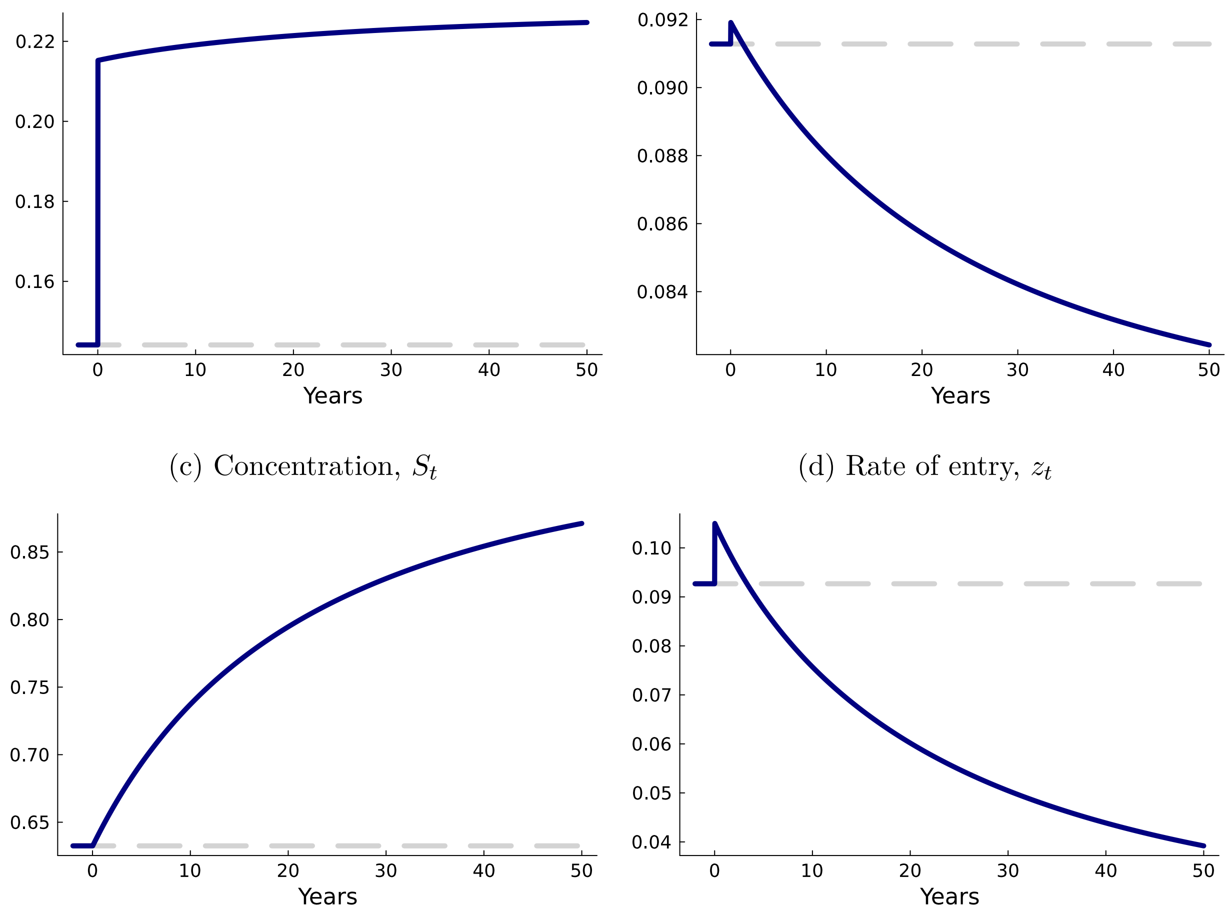
Notes: the figure shows output (consumption) along the transition towards the new balanced growth path. Parameters of the initial and new balanced growth path (BGP) are shown in Table 4. All parameter changes are introduced in period zero. Output is normalized to unity in year minus one for both BGPs.

Figure 9 shows the transitional dynamics for output (consumption) during the transition towards the new balanced growth path. Output along the initial balanced growth path is added for comparison in the dashed line. All parameter changes are introduced as one-time shocks in period zero. Output reacts positively on impact, resulting in a short-run boom in the economy relative to the initial balanced growth path. However, the growth burst is short-lived. Already after six years, the output path crosses with the trend of the initial balanced growth path and remains below. Figure 10 shows further equilibrium outcomes during the transition period. As the expansion R&D efficiency increases in period zero, the expansion R&D rates of both high- and low-productivity firms increase on impact. However, the increase is differently pronounced. Expansion R&D rates of high-productivity firms increase substantially, whereas the increase is muted for low-productivity firms. Expansion R&D rates continue to increase over time for high-productivity firms but gradually decrease for less productive ones. The increasing gap in expansion R&D rates results in a reallocation of market shares towards the high-productivity firms. The share of product lines operated by the high type, S_t, gradually increases over time. The bottom right panel shows the flow rate of entry z_t. The increase in the entry R&D efficiency increases z_t on impact; however, the increase is of short duration, as other forces, e.g., the rise in the expansion R&D efficiency and the fall in the step size of quality improvements, put downward pressure on firm entry.

Figure 10: Transitional dynamics for equilibrium outcomes

(a) Expansion R&D rate (high-type), x_t^h (b) Expansion R&D rate (low-type), x_t^ℓ

(c) Concentration, S_t (d) Rate of entry, z_t



Notes: the figure shows the response in equilibrium outcomes during the transition period towards the new balanced growth path. Parameters of the initial and new balanced growth path are shown in Table 4. All parameter changes are introduced in period zero. The dashed line indicates the initial balanced growth path.

That output increases in the short run relative to the initial balanced growth path in Figure 9 motivates the question of what the net effect on welfare is. To quantify the change in welfare, I compute the permanent consumption change (in percent) along the initial balanced growth path that makes the representative consumer as well off as with the obtained consumption stream over the transition towards the new balanced growth path. I find that welfare decreases by 4.95%. Hence, the long-run fall in output growth dominates the short-run increase. If one were to compare welfare of the two balanced growth paths without taking the transition nor any level effects into account, the consumption equivalent change ξ is determined by $\ln(1+\xi) = (g^{\text{new}} - g^{\text{initial}})/\rho$, where g^{new} and g^{initial} refer to the long-run aggregate growth rate in each balanced growth path. Given that g declines by 0.4 percentage points across the balanced growth paths and ρ equals 0.05, the welfare loss amounts to 7.69% ($\xi = -0.0769$). Therefore, the short-run output burst alleviates some of the fall in the long-run growth rate.

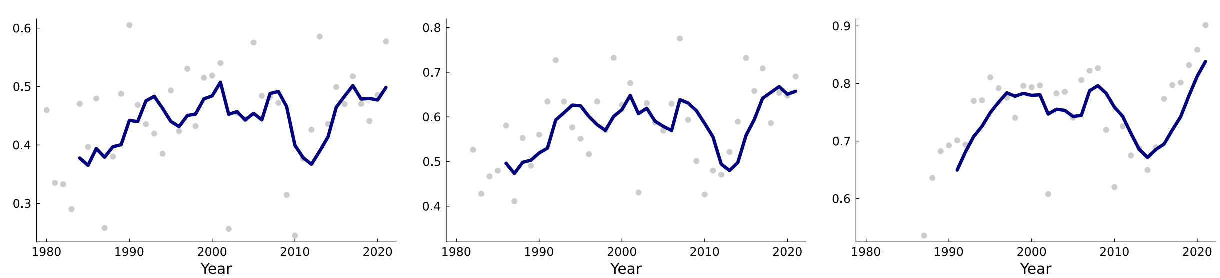
6 Robustness

6.1 Firm size trends in the U.S.

This section relates the documented trends in firm size for Sweden to the U.S. One advantage of the Swedish administrative data is that it contains information on firms' value added and the wage bill. Lacking such information for U.S. firms, I use the publicly available Census data (Business Dynamics Statistics). Figure 11 shows the average employment of U.S. firms aged three, five, and six to ten relative to firms of age zero in logs. Each time series is shown from its earliest availability (1980, 1982, and 1987, respectively) until the year 2021 in gray scattered dots. The blue line displays five-year moving averages. Starting with firms aged three, there is a clear upward trend visible in their relative size that begins around the late 1980s. Relative firm size continued to increase until the Great Recession, which caused a temporary drop and a quick rebound. Similar observations hold for firms aged five or six to ten. These time series start later and miss some of the rise in the relative firm size during the late 1980s (particularly the moving averages). Nevertheless, even for firms aged six to ten, the relative firm size steadily increased during the 1990s. The rise in the relative size of firms aged six to ten is even quantitatively comparable to Sweden. Whereas the relative wage bill of firms aged eight increased from 0.468 to 0.623 log points in Sweden, the relative employment of firms aged six to ten grew from 0.65 to 0.838 log points in the U.S. (starting and ending values of the moving average).

Figure 11: Firm size (employment) relative to entrants by age in the U.S.

(a) Age 3 (b) Age 5 (c) Age 6-10



Notes: the figures show average employment of firms aged three, five or six to ten relative to firms of age zero in logs. Data: U.S. Census Bureau - Center for Economic Studies - Business Dynamics Statistics (2021).

Hopenhayn, Neira and Singhania (2022) and Karahan, Pugsley and Şahin (2024) study firm size conditional on age in U.S. Census data, documenting no systematic trends over time. The difference to their papers is that they look at firm size conditional on age in levels, whereas I show firm size conditional on age *relative* to firms of age zero. What motivates the analysis relative to entrants? First, aggregate shocks that are not the focus of the long-run analysis, e.g., the China Shock or the Great Recession, affect firm size. Arguably, firm size (conditional on age) is more heavily affected in levels than relative to entrants by these shocks. One concrete example is the effect of the China Shock on firm size in the U.S. manufacturing sector. Whereas firm size conditional on age steadily declined, size relative to entrants remained much more stable. Second, firm size relative to entrants provides information about firm growth. In an endogenous growth model like the one in this paper, firm growth maps into aggregate productivity growth such that changes in the relative firm size contain valuable information about aggregate productivity growth.

6.2 Alternative parameterizations

In the main analysis, the parameter p^h capturing the share of high-productivity firms among entrants is set exogenously to a constant value of 0.5. This value was motivated by the fact that the interquartile range, the measure of industry concentration, captures size dispersion around the median firm. I relax this assumption in this section. In particular, I estimate the initial and new balanced growth path for alternative parameterizations of p^h and, as before, quantify the contributions of all parameters to the changes in the targeted moments each time.

Table 6 shows the estimated parameters in the initial and new balanced growth path for two alternative (exogenous) parameterizations of p^h, in particular $p^h = 0.4$ and $p^h = 0.6$. The results are similar to before. For both alternative parameterizations, the estimation asks for an increase in the expansion, internal, and entry R&D efficiencies and a fall in the step size of quality improvements. The only notable observation is that the estimation asks for a lower productivity gap in the initial balanced growth path when $p^h = 0.4$ than for $p^h = 0.6$. In other words, the estimation asks for more dispersion in the left part of the productivity distribution than in the right part. Likewise, the increase in the productivity gap across balanced growth paths is larger when p^h is set to 0.6. For p^h set to 0.4, the estimation even asks for a slight decrease in the productivity gap. In this case, the increase in concentration caused by the other parameter changes more than explains the observed increase such that no further increase in the productivity gap is needed. Therefore, the estimation indicates that dispersion primarily increases in the left part of the productivity distribution. This is an interesting observation, given that the previous literature emphasizes the role of so-called superstar firms in explaining recent macroeconomic trends.

Table 7 quantifies the contribution of each parameter to the changes in the targeted moments for both alternative parameterizations of p^h. As before, the decomposition follows eq. (27) and the notation Table 5. The results are quantitatively in line with the previous

Table 6: Estimation for alternative values of p^h

	Initial BGP		New BGP	
	Data	Model	Data	Model
$\boldsymbol{p^h = 0.4}$				
Moments				
Value added at age 8 relative to entry in logs	0.492	0.492	0.638	0.637
Wage bill at age 8 relative to entry in logs	0.468	0.468	0.623	0.623
Interquartile range of log value added	log(2)	log(2)	log(5)	log(5)
Entry rate in %	14.8	14.8	11	11
TFP growth g in %	1.5	1.5	1.1	1.1
Parameters				
ψ_x *Expansion R&D efficiency*		0.894		2.065
ψ_I *Internal R&D efficiency*		0.936		1.357
φ^h/φ^ℓ *Productivity gap*		1.017		1.014
ψ_z *Entry R&D efficiency*		3.786		6.329
λ *Step size of quality improvements*		1.051		1.035
$\boldsymbol{p^h = 0.6}$				
Moments				
Value added at age 8 relative to entry in logs	0.492	0.492	0.638	0.649
Wage bill at age 8 relative to entry in logs	0.468	0.468	0.623	0.632
Interquartile range of log value added	log(2)	log(2)	log(5)	log(5)
Entry rate in %	14.8	14.8	11	11
TFP growth g in %	1.5	1.5	1.1	1.1
Parameters				
ψ_x *Expansion R&D efficiency*		0.888		1.814
ψ_I *Internal R&D efficiency*		0.905		0.913
φ^h/φ^ℓ *Productivity gap*		1.022		1.038
ψ_z *Entry R&D efficiency*		3.792		6.522
λ *Step size of quality improvements*		1.051		1.038

parametrization of p^h. The increase in the expansion R&D efficiency and the fall in the step size of quality improvements drive the increase in firm size relative to entry (second and third column). At the same time, these two parameter changes explain most of the increase in concentration (fourth column), and the fall in firm entry (fifth column) and productivity growth (last column). As the increase in the productivity gap is more pronounced when p^h is set to 0.6, the productivity gap plays a more prominent role in explaining the targeted moments in this case.

Table 7: Decomposition for alternative values of p^h

	VA	wL	σVA	$\frac{z}{M}$	g
$\boldsymbol{p^h = 0.4}$					
Expansion R&D efficiency $\psi_x \uparrow$	380.3	364.7	108.4	184.4	-39.6
Internal R&D efficiency $\psi_I \uparrow$	3.7	-2.2	-4.7	-31.2	-54.8
Productivity gap $\varphi^h/\varphi^\ell \downarrow$	-3.2	-2.7	-2.8	-7.5	-1.4
Entry R&D efficiency $\psi_z \uparrow$	-126.5	-115.4	-40.3	-346.0	-94.4
Step size of quality improvements $\lambda \downarrow$	8.5	12.4	33.1	194.8	201.7
$\boldsymbol{p^h = 0.6}$					
Expansion R&D efficiency $\psi_x \uparrow$	281.0	275.0	95.7	128.0	-33.9
Internal R&D efficiency $\psi_I \uparrow$	0.1	-0.1	-0.1	-0.6	-1.1
Productivity gap $\varphi^h/\varphi^\ell \uparrow$	24.3	21.3	25.5	51.5	9.2
Entry R&D efficiency $\psi_z \uparrow$	-120.5	-112.0	-49.3	-347.8	-103.6
Step size of quality improvements $\lambda \downarrow$	6.5	9.7	30.3	147.6	163.5

Notes: the table shows the contribution of each parameter change (rows) to the changes in the five moments (columns) across the intial and new balanced growth path (BGP) in percent. The contribution of each parameter change is computed using eq. (27). Moments and parameter values of the initial and new BGP are shown in Table 6. VA denotes the difference in average log value added between firms aged eight and entrants, wL the difference in the average log wage bill between firms aged eight and entrants, σVA the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate.

6.3 Firm size growth and selection

The analysis so far has remained silent on whether the increase in firm size relative to entry is caused by changes in firm growth or the selection of (surviving) firms. This section disentangles both margins with the help of the structural model. In particular, I show how firm size growth and the selection of surviving firms change across the balanced growth paths, giving rise to the targeted increase in firm size relative to entry.

Table 8: Firm size growth and selection

	Initial BGP	New BGP
Expansion R&D rate (high-prod.) x^h	0.144	0.228
Expansion R&D rate (low-prod.) x^ℓ	0.091	0.08
Share (high-prod.) among surviving firms	0.543	0.619
Growth cond. on survival (high-prod.)	0.577	0.842
Growth cond. on survival (low-prod)	0.391	0.307

Notes: the table shows the expansion R&D rates of high- and low-productivity firms (first two rows), the share of high-productivity firms among firms alive at age eight (third row) and firm size growth (value added) over the first eight years for high- and low productivity firms in logs. Parameter values of the initial and new balanced growth path (BGP) are shown in Table 4.

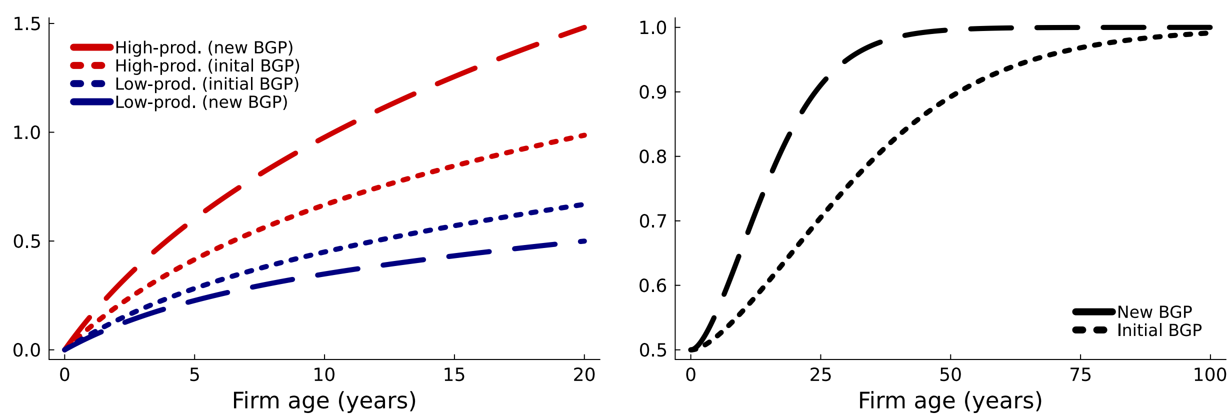
Firm size and survival are directly related. Both depend on the number of products a firm produces. Hence, changes in the firms' expansion R&D rates directly affect their growth and survival. Table 8 shows the expansion R&D rates of high and low productivity firms in the initial and new balanced growth path. The expansion R&D rate of high-productivity firms increases from 0.144 to 0.228, whereas the one of low-productivity firms decreases from 0.091 to 0.08. As a consequence, size growth (value added) conditional on survival over the first eight years increases for high-productivity firms from 0.577 to 0.842 log points. Size growth of low-productivity firms decreases from 0.391 to 0.307 log points in the new balanced growth path. The increase (decrease) in expansion R&D rates of high (low) productivity firms increases (decreases) their survival probabilities. Hence, the share of high-productivity firms among firms that survive up to age eight, $s^h(8)$, increases from 0.543 to 0.619.

Eq. (24) shows that firm size relative to entry is equal to firm size growth conditional on survival of high- and low-productivity firms weighted by their respective shares among the surviving firms (the last term in the equation disappears when firm size is measured using value added). Hence, multiplying the last two rows in Table 8 by the share of high- and low-type firms among firms aged eight reported in the third last row gives firm size (value added) relative to entry, as targeted in the estimation of the initial and new balanced growth path. Hence, firm size relative to entry increases across the balanced growth paths due to both changes in firm growth conditional on survival and the selection of surviving firms. More productive firms grow faster conditional on survival and represent a larger share among surviving firms. Both increase firm size relative to entry.

Figure 12: Firm size growth and selection

(a) Firm size growth

(b) Selection



Notes: the left figure shows firm size (value added) relative to entry (in logs) for high and low productivity firms as a function of firm age. The right figure shows the share of high productivity firms among surviving firms as a function of firm age.

Figure 12 makes this point more generally, showing firm size growth (value added) conditional on survival for high- and low-productivity firms in the left panel and the share of high-productivity firms among surviving firms, s^h, in the right panel as a function of firm age. At age eight, the values correspond to the ones shown in Table 8. The increase (decrease) in the expansion R&D rates of high (low) productivity firms increases (decreases) firm growth over the entire life cycle. The share of high-productivity firms among survivors in the right panel starts at 0.5 at age zero (the share of high-type firms among entrants p^h) and eventually converges to one. The share converges to one as high-productivity firms are more likely to survive than less-productive ones due to their higher expansion R&D rates. In other words, despite only a few firms surviving until old age, those that do are more likely to be of the high-productivity type. As the gap in the expansion R&D rates increases across balanced growth paths, convergence happens faster, i.e., the share of high-productivity firms among survivors increases for any firm age.

6.4 Discussion on markups

Evidence on markups for the Swedish economy is scarce. Sandström (2020) provides an estimate of the aggregate markup using, as I do, administrative data on the balance sheets of Swedish firms. The author restricts to firms with at least ten employees. As I do not impose such restriction in my data, I view the markup estimates in Sandström (2020) as an upper bound for my analysis, as markups tend to increase in firm size (Edmond, Midrigan and Xu, 2023). The author reports an estimate of about 15% for the (sales-weighted) aggregate markup and 5% for the unweighted one. Both estimates are stable, displaying no systematic trend over time. These estimates are lower than common markup estimates for the U.S. (De Loecker, Eeckhout and Unger, 2020); however, a direct comparison is difficult due to differences in, e.g., the data coverage or the institutional setting.

In this model, the (cost-weighted) markup in the estimated initial balanced growth path stands at 7.1%, well within the estimates provided in Sandström (2020). The aggregate markup even declines slightly in the new balanced growth path. Despite high-productivity firms with relatively high markups capturing additional market shares in the new balanced growth path, the fall in the step size of quality improvements puts downward pressure on markups. One could extend the theory in this paper to allow incumbent firms to distance competitors within product markets without positive effects on productivity growth through, e.g., a decline in knowledge diffusion as in Akcigit and Ates (2023) to generate increasing markups while productivity growth falls. Increasing markups at the product or firm level, however, imply a rise in the wedge between growth in firms' value added and the wage bill. If anything, this wedge slightly decreases in my data as shown in Table 4. This and the fact that Sandström (2020) does not find systematic changes in the aggregate markup motivates the exclusion of the latter from the main analysis in this paper.

7 Conclusion

This paper documents a novel observation about the dynamics of firm size. Firm size relative to entry has systematically increased for firms established around the turn of the millennium. This paper uses this novel observation as a firm-level moment to test different explanations highlighted in the literature as the cause behind macroeconomic trends observed over the last 30 years, such as a rise in industry concentration and a fall in firm entry and aggregate productivity growth.

To this end, I leverage an endogenous growth model with rich firm dynamics that accommodates many of the highlighted forces to explain the increase in firm size relative to entry and the macroeconomic trends jointly. Estimating the model along separate balanced growth paths shows that of the highlighted forces in the literature, a decrease in the firm expansion costs and innovations becoming more incremental quantitatively account for the increase in firm size relative to entry. At the same time, these two forces explain most of the rise in industry concentration and the fall in firm entry and productivity growth. Other forces, such as increasing productivity dispersion, lower costs of distancing competitors within product markets, or rising entry costs, do not significantly affect firm size relative to entry or cause counterfactual changes in the macroeconomy when explaining the firm-level trends quantitatively.

Why have firm expansion costs fallen and innovations become more incremental? Potentially, several forces have contributed to the decline in firm expansion costs. One standard guess is that transportation costs have decreased. Another conjecture holds that new technologies allow firms to operate and supervise more production units at lower costs (Aghion, Bergeaud, Boppart, Klenow and Li, 2023; Hsieh and Rossi-Hansberg, 2023). Relatedly, both the computer and the internet were revolutionary inventions that broke out in the 1990s. The IT revolution was so groundbreaking that subsequent innovations found it difficult to keep up. Future research could provide direct empirical evidence on the fall in firm expansion costs and the quality of innovations. Understanding the origins of the highlighted forces appears key, given their importance in explaining the macroeconomic trends.

The model has further rich predictions that can be tested in microdata. Two of them are that more productive firms grow faster than they used to (less productive ones slower) and that the share of the former among surviving firms has increased. Further, the model is intentionally kept comparable to other models in the literature to ensure a fair comparison of the different mechanisms. There are numerous ways to extend the model, e.g., introduce more firm-productivity types. I leave these interesting endeavors for future research.

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Online Appendix for

"Recent Changes in Firm Dynamics and the Nature of Macroeconomic Trends"

[FOR ONLINE PUBLICATION]

A Data

The main data set, *Företagens Ekonomi* (FEK), covers information from balance sheets and profit and loss statements for the universe of Swedish firms. From this data, I obtain the main variables of interest, namely value added (*Förädlingsvarde*, variable name: *Foradlingsvarde*) and the wage bill (*Lönekostnader*, variable name: *KonstnaderForLonerOchAndraErsattn*). In the FEK code-book by Statistics Sweden (SCB), these variables are defined as follows.[17] SCB defines value added as a firm's production value minus costs for purchased goods and services used as inputs in production (salaries and social security contributions not included). SCB's measure of value added captures the firm's contribution to the gross domestic product. The measure of the firm's wage bill includes salaries and other remuneration, including severance pay. As described in the main text, I focus on firms in the private sector. These firms have a legal type (variable name: *JurForm*) less than 50 or equal to 96.

To measure industry concentration, I use the industry classification (SNI codes) provided by SCB. The industry classification changed twice between 1997 and 2017, once in 2002 and once in 2007. The changes in industry classification mainly affect the more detailed codes at the five-digit level, whereas I use the ones at the two-digit level. Nevertheless, to ensure a consistent industry classification over time I proceed as follows. During the year of the

[17] `https://www.scb.se/contentassets/9dd20ce462644cc19f6f04eb2edbbe28/nv0109_kd_slut2022_v1_20240510.pdf`, accessed 26.03.2025.

change, I observe both the old and the new industry classifications. For the firms present in the data in the year of the classification change, extending the new industry classification further back in time before the change is straightforward. This way, the industry codes of almost all firms are updated. A firm might be in the data before and after the classification change but not for the year of the change. For these firms, the above method does not work. If the firm appears in the data one year after the classification change, I use the observed classification after the change to update the classification before the change. For firms that are absent for several years around the year of change, I use industry mappings provided by Statistics Sweden. These mappings do not always provide a 1:1 mapping between industries before and after the classification change, so I use the most common transitions for the m:m mappings.

One concern is that changes in the firm structure, e.g., when firms merge with other firms, change the firm ID. To address this concern, I impute changes in firm IDs using worker flows between firms. The auxiliary data set *Registerbaserad Arbetsmarknadsstatistik* (RAMS) contains the universe of employer-employee matches. I impute changes in the firm ID of firms with at least five employees as follows: if more than 50% of the workforce of firm A in year t makes up for more than 50% of the workforce of firm B in year $t+1$, I substitute firm B's firm ID by firm A's firm ID following $t+1$. The empirical results remain unchanged when excluding firms for whom the imputed firm ID differs from the observed firm ID.

Other data sources

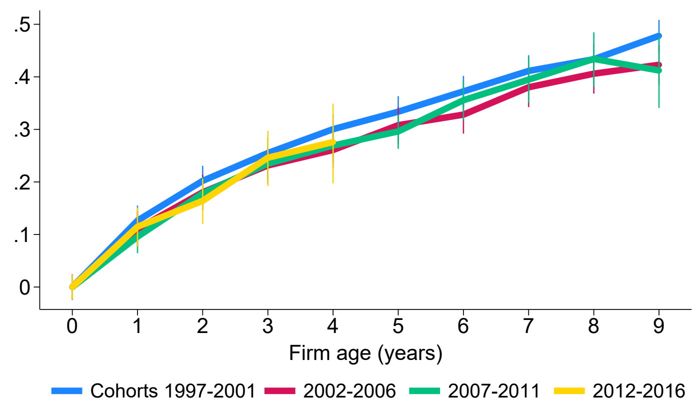
- U.S. Census Bureau - Center for Economic Studies - Business Dynamics Statistics (2021). Accessed July 13, 2024.
- Federal Reserve Economic Data (FRED): Total Factor Productivity at Constant National Prices for Sweden (series RTFPNASEA632NRUG). Accessed Tuesday, January 30, 2024.

B Trends in firm size

B.1 Firm-size trends in Sweden

This section documents additional trends in firm size in the Swedish economy. Figure A-1 shows the average physical capital stock relative to entry by cohort (in logs). The capital stock is defined as fixed assets minus depreciation. The different cohort trajectories overlap, indicating no systematic difference in firms' capital accumulation over time. This suggests that the increase in firm size relative to entry documented in the main text is not driven by capital deepening.

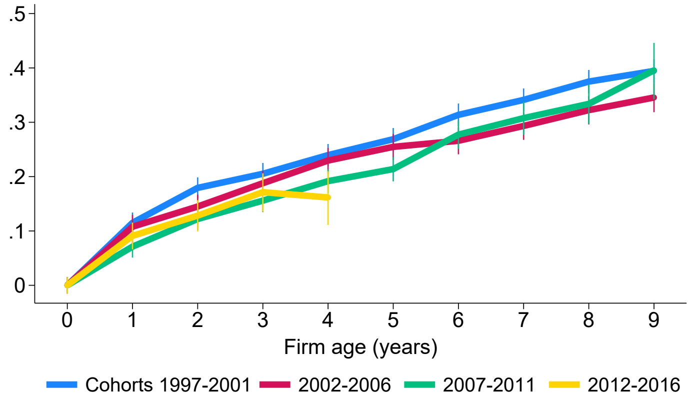
Figure A-1: Capital stock relative to entry



Notes: the figure shows the average capital stock of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. The cohort 2004 is excluded as explained in the main text. 95% confidence bands are included.

This picture does not change even when including intangible assets in the measure of the capital stock. Figure A-2 repeats the previous graph for the measure of capital that includes firms' (book value of) intangible capital. The trajectories become slightly more dispersed compared to Figure A-1; however, there is no systematic trend across cohorts visible.

Figure A-2: Capital stock (incl. intangibles) relative to entry

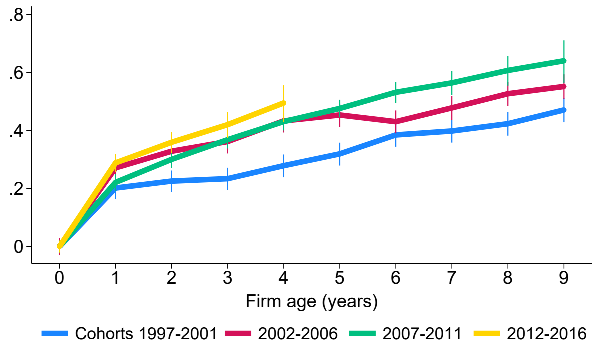


Notes: the figure shows the average capital stock (including intangibles) of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. The cohort 2004 is excluded as explained in the main text. Intangible capital is filtered at its top 0.5% tail. 95% confidence bands are included.

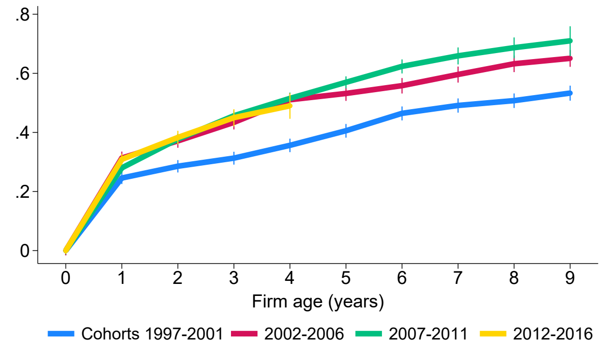
Next, I show the size trajectories of value added and the wage bill by sector. The left column in Figure A-3 is restricted to firms in the goods sector, and the right column to firms in the services sector. The trends are visible in both sectors, particularly in the services sector. That the trends are particularly clear for services firms whose products generally are locally provided and consumed suggests that the trends in firm size are not driven by foreign goods demand.

Figure A-3: Firm-size relative to entry (by sector)

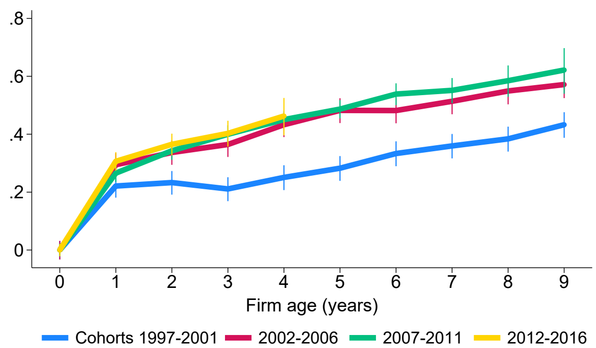
(a) Value added (Goods)



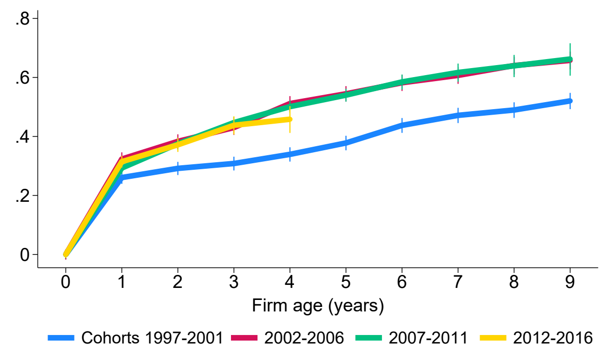
(b) Value added (Services)



(c) Wage bill (Goods)



(d) Wage bill (Services)



Notes: the figure shows the average value added and wage bill of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. The left column is restricted to firms in the goods sector and the right column to firms in the services sector. 95% confidence bands are included.

C Model

C.1 Solving the dynamic firm problem

The HJB for a high productivity-type firm h reads[18]

$$
\begin{aligned}
&r_t V_t^h(n,[\mu_i],S_t) - \dot{V}_t^h(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^h\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \Bigg\{ \sum_{k=1}^{n} I_k \Big[V_t^h\left(n,\left[[\mu_i]_{i\neq k},\mu_k\times\lambda\right],S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^h\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) + (1-S_t) V_t^h\Big(n+1,\Big[[\mu_i],\lambda\times\varphi^h/\varphi^l\Big],S_t\Big) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad - w_t \left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^\zeta + \frac{1}{\psi_x}(x_k)^\zeta \right] \Bigg\}
\end{aligned}
$$

The HBJ for a low productivity-type firm l reads

$$
\begin{aligned}
&r_t V_t^l(n,[\mu_i],S_t) - \dot{V}_t^l(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^l\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \Bigg\{ \sum_{k=1}^{n} I_k \Big[V_t^l\left(n,\left[[\mu_i]_{i\neq k},\mu_k\times\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^l\Big(n+1,\Big[[\mu_i],\lambda\times\varphi^l/\varphi^h\Big],S_t\Big) + (1-S_t) V_t^l\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad - w_t \left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^\zeta + \frac{1}{\psi_x}(x_k)^\zeta \right] \Bigg\}.
\end{aligned}
$$

I solve for the value function of a high-type firm, however the steps for the low-type firm are equivalent. For clarity, I suppress the dependence of the value function on S_t in the following. Guess that the value function of the firm consists of a component that is common to all lines and a line-specific component

$$
V_t^h(n,[\mu_i]) = V_{t,P}^h(n) + \sum_{k=1}^{n} V_{t,M}^h(\mu_k).
$$

[18] The notation follows Peters (2020).

Substituting the guess into the HJB, $V^h_{t,P}(n)$ and $V^h_{t,M}(\mu_k)$ solve the following differential equations

$$\begin{aligned} r_t V^h_{t,M}(\mu_i) - \dot{V}^h_{t,M}(\mu_i) =& \pi(\mu_i) - \tau_t V^h_{t,M}(\mu_i) \\ &+ \max_{I_i} \left\{ I_i \left[V^h_{t,M}(\mu_i \times \lambda) - V^h_{t,M}(\mu_i) \right] - w_t \mu_i^{-1} \frac{1}{\psi_I} (I_i)^\zeta \right\} \end{aligned} \tag{A-1}$$

and

$$\begin{aligned} & r_t V^h_{t,P}(n) - \dot{V}^h_{t,P}(n) = \sum_{k=1}^{n} \tau_t \left[V^h_{t,P}(n-1) - V^h_{t,P}(n) \right] \\ & + \max_{[x_k]} \left\{ \sum_{k=1}^{n} x_k \left[V^h_{t,P}(n+1) - V^h_{t,P}(n) + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) \right] - w_t \frac{1}{\psi_x} (x_k)^\zeta \right\}. \end{aligned} \tag{A-2}$$

Assume that in steady-state $V^h_{t,P}$ and $V^h_{t,M}$ grow at the constant rate g. Using this guess in eq. (A-1) and following Peters (2020), we obtain for $V^h_{t,M}(\mu_i)$

$$V^h_{t,M}(\mu_i) = \frac{\pi(\mu_i) + \frac{\zeta-1}{\psi_I}(I_i)^\zeta w_t \mu_i^{-1}}{\rho+\tau},$$

where I_i solves

$$I_i = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta-1}{\psi_I}(I_i)^\zeta \right) \left(1 - \frac{1}{\lambda}\right) \frac{\psi_I}{\zeta(\rho+\tau)} \right)^{\frac{1}{\zeta-1}}. \tag{A-3}$$

Eq. (A-3) shows that internal innovation rates I_i are time invariant, and independent of the product line and the productivity type of the firm, $I \equiv I^h = I^l$.

With this at hand, we can turn back to the differential equation for $V^h_{t,P}(n)$ in eq. (A-2). In addition to the guess that $V^h_{t,P}(n)$ grows at rate g, conjecture that $V^h_{t,P}(n) = n \times v^h_t$. Combined with the Euler we get

$$(\rho+\tau) n v^h_t = \max_{[x_k]} \left\{ \sum_{k=1}^{n} x_k \left[v^h_t + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) \right] - w_t \frac{1}{\psi_x}(x_k)^\zeta \right\}. \tag{A-4}$$

The optimality condition for x_k is given by

$$v^h_t + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) = w_t \frac{\zeta}{\psi_x}(x_k)^{\zeta-1}. \tag{A-5}$$

Several observations are noteworthy. First, eq. (A-5) shows that optimal expansion rates are independent of quality and productivity gaps in line k. We can hence drop the item indexation: $x_k = x^d$, where $d \in \{h, \ell\}$. Second, $v_t, V^h_{t,M}, w_t$ all grow at the same rate g, which implies that expansion rates are constant over time. We can hence write eq. (A-4) as

$$v^h_t = \frac{1}{(\rho+\tau)} \frac{\zeta-1}{\psi_x} (x^h)^\zeta w_t.$$

Gathering all terms, the value function is given by

$$\begin{aligned} V^h_t(n, [\mu_i]) =& V^h_{t,P}(n) + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\ =& n v^h_t + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\ =& n \frac{1}{(\rho+\tau)} \frac{\zeta-1}{\psi_x} (x^h)^\zeta w_t + \sum_{k=1}^{n} \frac{\pi(\mu_k) + \frac{\zeta-1}{\psi_I} I^\zeta w_t \mu_k^{-1}}{\rho+\tau}, \end{aligned} \tag{A-6}$$

which is the expression for the value function stated in the main text, Proposition 1.

Using the expression for v^h_t, write the optimality condition in eq. (A-5) as

$$\begin{aligned} \frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t \left(\frac{Y_t}{w_t} \left(1 - \frac{1}{\lambda}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \right) + (1-S_t) \left(\frac{Y_t}{w_t} \left(1 - \frac{\varphi^l}{\varphi^h} \frac{1}{\lambda}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \frac{\varphi^l}{\varphi^h} \right) \\ = (\rho+\tau) \frac{\zeta}{\psi_x} (x^h)^{\zeta-1}. \end{aligned}$$

Following the same steps for low-productivity firms, we obtain the optimality condition

$$\begin{aligned} \frac{\zeta-1}{\psi_x}(x^l)^\zeta + S_t \left(\frac{Y_t}{w_t} \left(1 - \frac{1}{\lambda} \frac{\varphi^h}{\varphi^l}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \frac{\varphi^h}{\varphi^l} \right) + (1-S_t) \left(\frac{Y_t}{w_t} \left(1 - \frac{1}{\lambda}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \right) \\ = (\rho+\tau) \frac{\zeta}{\psi_x} (x^l)^{\zeta-1}. \end{aligned}$$

Lastly, I prove that more productive firms choose higher expansion R&D rates, i.e., $x^h > x^l$. Intuitively, the proof shows that an increase in productivity raises the stream of profits in a product line. The continuation value and the marginal cost of expansion R&D have to rise for the optimality condition of the expansion R&D rate to hold, implying that the expansion R&D rate increases. First note that product markups are increasing in firm productivity, as shown in eq. (3). Next, I totally differentiate the optimality condition for the expansion

R&D rate and show that the expansion R&D rate is increasing in the markup.

Write the optimality condition for expansion R&D as[19]

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^h}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^h}\right)+(1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^l}\right)$$
$$=(\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1},$$

where μ^h and μ^l denote the initial markup charged when facing a high- and low-productivity firm. Totally differentiate with respect to markups and the expansion R&D rate

$$S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left((\rho+\tau)(x^h)^{\zeta-2}-(x^h)^{\zeta-1}\right)dx^h,$$

where $d\mu^h = d\mu^l \equiv d\mu$. Since $x^h > 0$, the above can be rearranged to

$$\frac{1}{(x^h)^{\zeta-2}}\left[S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)\right]d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left(\rho+\tau-x^h\right)dx^h.$$

The left hand side captures the effect of changes in markups on profits. The right hand side captures the effect of changes in the expansion R&D rate on the continuation value and marginal costs of expansion R&D. From eq. (A-3) we know that $\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta > 0$, otherwise the optimal internal R&D rate is negative. For a stationary firm size distribution, we must further have $\tau > x^h$. From this, it follows that $dx^h/d\mu > 0$, which concludes the proof.

C.2 Joint distribution of quality and productivity gaps

I characterize the two-dimensional distribution of quality and productivity gaps along the BGP as a function of firm policies. This allows for optimal policies and the distribution to be solved jointly. I solve for the steady state distribution over quality and productivity gaps by setting the differential equations characterizing the law-of-motion in eq. (10) and (11) equal to zero. From this, one obtains the stationary mass of product lines with quality gap

[19] This uses the optimality condition of the high-type firm but the one of the low type works equivalently.

λ^Δ and productivity gap φ^i/φ^j

$$\begin{aligned}
\nu\left(\Delta, \frac{\varphi^l}{\varphi^h}\right) &= \left(\frac{I}{I+\tau}\right)^\Delta \frac{(1-S)x^l S + z(1-p^h)S}{I} \\
\nu\left(\Delta, \frac{\varphi^l}{\varphi^l}\right) &= \left(\frac{I}{I+\tau}\right)^\Delta \frac{(1-S)x^l(1-S) + z(1-p^h)(1-S)}{I} \\
\nu\left(\Delta, \frac{\varphi^h}{\varphi^h}\right) &= \left(\frac{I}{I+\tau}\right)^\Delta \frac{Sx^h S + zp^h S}{I} \\
\nu\left(\Delta, \frac{\varphi^h}{\varphi^l}\right) &= \left(\frac{I}{I+\tau}\right)^\Delta \frac{Sx^h(1-S) + zp^h(1-S)}{I}.
\end{aligned}$$

Summing over all Δ for a given productivity gap gives $S_{\varphi^l,\varphi^h}, S_{\varphi^l,\varphi^l}, S_{\varphi^h,\varphi^h}, S_{\varphi^h,\varphi^l}$ as stated in Proposition 1 in the main text. It follows that

$$\begin{aligned}
\Pr\left(\Delta \leq d, \frac{\varphi^l}{\varphi^h}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right) \\
\Pr\left(\Delta \leq d, \frac{\varphi^l}{\varphi^l}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^l}{\varphi^l}\right) = S_{\varphi^l,\varphi^l}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right) \\
\Pr\left(\Delta \leq d, \frac{\varphi^h}{\varphi^h}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^h}{\varphi^h}\right) = S_{\varphi^h,\varphi^h}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right) \\
\Pr\left(\Delta \leq d, \frac{\varphi^h}{\varphi^l}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^h}{\varphi^l}\right) = S_{\varphi^h,\varphi^l}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right).
\end{aligned}$$

Focusing on product lines where a low-productivity incumbent faces a high-productivity second-best firm:

$$P\left(\Delta \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - e^{-d[\ln(I+\tau) - \ln I]}\right)$$

or

$$P\left(\ln\left(\lambda^\Delta\right) \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - e^{-\frac{\ln(I+\tau) - \ln I}{\ln(\lambda)} d}\right).$$

Conditional on the productivity gap, $\ln\left(\lambda^\Delta\right)$ is exponentially distributed with parameter

$\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Further

$$P\left(\lambda^{\Delta} \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - d^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter $\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Denote $\theta = \frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. We then have

$$P\left(\lambda^{\Delta} \leq m, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - m^{-\theta}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter θ. As in Peters (2020), θ is affected by the rate of internal R&D I relative to creative destruction τ. The higher the rate of internal R&D, the more mass is in the tail of the quality gap distribution. The difference to Peters (2020) is that, in this model, quality gaps *conditional* on the productivity gap are Pareto distributed.

After repeating the same steps for lines with different productivity gaps, we obtain the aggregate labor income share as follows[20]

$$\begin{aligned}\Lambda &= \sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int_1^{\infty}\frac{1}{\varphi_k/\varphi_n}\frac{1}{m}S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm \\ &= \frac{\theta}{\theta+1}\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n}.\end{aligned}$$

The TFP misallocation statistic $\mathcal{M}$ is then given by

$$\begin{aligned}\mathcal{M} &= \frac{e^{\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int\left[\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\frac{1}{m}\right)S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}\right]dm}}{\Lambda} \\ &= \frac{e^{\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\left[S_{\varphi_k,\varphi_n}\left(\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\right)-\frac{1}{\theta}\right)\right]}}{\Lambda},\end{aligned}$$

where I have made use of

$$\int_1^{\infty}\ln\left(\frac{1}{m}\right)S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm = \left[\frac{\theta\ln(m)+1}{\theta m^{\theta}} + C\right]_1^{\infty} = -\frac{1}{\theta}.$$

Alternatively note that this expression is equal to $-S_{\varphi_k,\varphi_n}E[\ln(\lambda^{\Delta})|\varphi_k,\varphi_n]$. I have shown

[20] For the derivation, I assume a continuous distribution of quality gaps.

above that $\ln(\lambda^{\Delta})$ conditional on the productivity gap is exponentially distributed with parameter θ. From the characteristics of an exponential distribution, its expected value is $1/\theta$.

The aggregate markup is then given by

$$
\begin{aligned}
E[\mu] = & \sum_{k\in\{h,l\}} \sum_{n\in\{h,l\}} \int_1^{\infty} \frac{\varphi_k}{\varphi_n} m S_{\varphi_k,\varphi_n} \theta m^{-(\theta+1)} dm \\
= & \frac{\theta}{\theta-1} \sum_{k\in\{h,l\}} \sum_{n\in\{h,l\}} \frac{\varphi_k}{\varphi_n} S_{\varphi_k,\varphi_n}.
\end{aligned}
$$

This concludes the derivations of the aggregate labor income share, TFP misallocation statistic and aggregate markup. In Peters (2020), the step size of innovation and the rate of creative destruction relative to internal R&D, captured by θ, fully characterize the aggregate labor income share, the misallocation measure, and the aggregate markup. In this model, these statistics further depend on the size and distribution of the productivity gaps. For example, a rise in the productivity gap or a reallocation of sales shares towards high-productivity firms lowers the aggregate labor income share and raises the markup, ceteris paribus.

C.3 Deriving the steady-state growth rate of aggregate variables

The growth rate of Q_t determines the growth rate of aggregate variables

$$
g = \frac{\dot{Q}_t}{Q_t} = \frac{\partial \ln(Q_t)}{\partial t}.
$$

Quality of a product in a given product line increases through internal R&D, expansion R&D or firm entry. For the growth rate of Q_t over a discrete time interval Δ, we have

$$
\ln(Q_{t+\Delta}) = \int_0^1 \Big[(\Delta I + \Delta S x^h + \Delta(1-S)x^l + \Delta z)\ln(\lambda) + \ln(q_{t,i})\Big]\, di
$$

so that

$$
\frac{\ln(Q_{t+\Delta}) - \ln(Q_t)}{\Delta} = \Big(I + Sx^h + (1-S)x^l + z\Big)\ln(\lambda).
$$

For $\Delta \to 0$, $g = \Big(I + Sx^h + (1-S)x^l + z\Big)\ln(\lambda)$ as stated in Proposition 1.

C.4 Solving for the steady state equilibrium

In the model, there are the seven unknown variables $x^h, x^l, I, z, \tau, \frac{Y_t}{w_t}, S$ and the markup distribution $\nu()$ in seven equations plus the system of differential equations characterizing $\nu()$.

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta-1}{\psi_I}I^\zeta\right)\left(1-\frac{1}{\lambda}\right)\frac{\psi_I}{\zeta(\rho+\tau)}\right)^{\frac{1}{\zeta-1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\right)+(1-S)\left(\frac{Y_t}{w_t}\left(1-\frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\frac{\varphi^l}{\varphi^h}\right)$$
$$= (\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta-1}{\psi_x}(x^l)^\zeta + S\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\frac{\varphi^h}{\varphi^l}\right)+(1-S)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\right)$$
$$= (\rho+\tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta-1}$$

Free entry condition

$$p^h\Big(SV_t^h(1,\lambda)+(1-S)V_t^h(1,\lambda\times\varphi^h/\varphi^l)\Big)+(1-p^h)\Big(SV_t^l(1,\lambda\times\varphi^l/\varphi^h)+(1-S)V_t^l(1,\lambda)\Big)=\frac{1}{\psi_z}w_t,$$

where

$$V_t^d(1,\mu) = \frac{1}{(\rho+\tau)}\frac{\zeta-1}{\psi_x}(x^d)^\zeta w_t + \frac{Y_t\left(1-\frac{1}{\mu}\right)+\frac{\zeta-1}{\psi_I}I^\zeta w_t\mu^{-1}}{\rho+\tau}$$

Labor market clearing condition

$$1=\frac{Y_t}{w_t}\sum_{\frac{\varphi_j}{\varphi_{j'}}}\sum_i\frac{1}{\lambda^i\frac{\varphi_j}{\varphi_{j'}}}\nu\left(i,\frac{\varphi_j}{\varphi_{j'}}\right)+\frac{1}{\psi_I}I^\zeta\sum_{\frac{\varphi_j}{\varphi_{j'}}}\sum_i\frac{1}{\lambda^i\frac{\varphi_j}{\varphi_{j'}}}\nu\left(i,\frac{\varphi_j}{\varphi_{j'}}\right)+S\frac{1}{\psi_x}(x^h)^\zeta+(1-S)\frac{1}{\psi_x}(x^l)^\zeta+\frac{z}{\psi_z}$$

Creative destruction

$$\tau = z + Sx^h + (1-S)x^l$$

Share of high productivity type

$$S = \sum_{i=1}^{\infty} \left[\nu \left(i, \frac{\varphi^h}{\varphi^h} \right) + \nu \left(i, \frac{\varphi^h}{\varphi^l} \right) \right],$$

where ν, the stationary distribution of quality and productivity gaps, is characterized by

$$0 = \dot{\nu} \left(\Delta, \frac{\varphi_j}{\varphi_{j'}} \right) = I\nu \left(\Delta - 1, \frac{\varphi_j}{\varphi_{j'}} \right) - \nu \left(\Delta, \frac{\varphi_j}{\varphi_{j'}} \right) (I + \tau) \quad \text{for} \quad \Delta \geq 2$$

and for the case of a unitary quality gap

$$\begin{aligned}
0 = \dot{\nu} \left(1, \frac{\varphi^l}{\varphi^h} \right) &= (1-S)x^l S + z_t(1-p^h)S - \nu \left(1, \frac{\varphi^l}{\varphi^h} \right) (I + \tau) \\
0 = \dot{\nu} \left(1, \frac{\varphi^l}{\varphi^l} \right) &= (1-S)x^l(1-S) + z_t(1-p^h)(1-S) - \nu \left(1, \frac{\varphi^l}{\varphi^l} \right) (I + \tau) \\
0 = \dot{\nu} \left(1, \frac{\varphi^h}{\varphi^h} \right) &= Sx^h S + z_t p^h S - \nu \left(1, \frac{\varphi^h}{\varphi^h} \right) (I + \tau) \\
0 = \dot{\nu} \left(1, \frac{\varphi^h}{\varphi^l} \right) &= Sx^h(1-S) + z_t p^h(1-S) - \nu \left(1, \frac{\varphi^h}{\varphi^l} \right) (I + \tau).
\end{aligned}$$

To simplify the system of equations, first rewrite the rate of creative destruction

$$z = (\tau - Sx^h - (1-S)x^l)$$

such that z can be substituted out from the remaining equations. Second, based on Proposition 1, we know

$$S = S_{\varphi^h, \varphi^h} + S_{\varphi^h, \varphi^l} = \frac{Sx^h + zp^h}{\tau}.$$

Third, the optimality conditions for expansion rates (multiplied by p^h and $(1-p^h)$) and the free entry condition together imply

$$\frac{1}{\psi_x} p^h (x^h)^{\zeta - 1} + \frac{1}{\psi_x} (1-p^h)(x^l)^{\zeta - 1} = \frac{1}{\psi_z \zeta}.$$

The system of equilibrium conditions can hence be reduced to:

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} \psi_I - (\zeta - 1) I^\zeta \right) \frac{\left(1 - \frac{1}{\lambda}\right)}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^h)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^\zeta \lambda^{-1}\frac{1}{\psi_I}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + (\zeta - 1)I^\zeta \lambda^{-1}\frac{\varphi^l}{\psi_I \varphi^h}\right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta - 1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^l)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + (\zeta - 1)I^\zeta \lambda^{-1}\frac{\varphi^h}{\psi_I \varphi^l}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^\zeta \lambda^{-1}\frac{1}{\psi_I}\right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta - 1}$$

Free entry

$$p^h \frac{(x^h)^{\zeta - 1}}{\psi_x} + (1 - p^h)\frac{(x^l)^{\zeta - 1}}{\psi_x} = \frac{1}{\psi_z \zeta}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t}\Lambda + \Lambda_I + S\frac{1}{\psi_x}(x^h)^\zeta + (1 - S)\frac{1}{\psi_x}(x^l)^\zeta + \frac{\tau - Sx^h - (1 - S)x^l}{\psi_z},$$

where

$$\Lambda = \frac{\theta}{\theta + 1}\sum_{k \in \{h,l\}}\sum_{n \in \{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n}$$
$$\Lambda_I = \frac{1}{\psi_I}I^\zeta \frac{\theta}{\theta + 1}\sum_{k \in \{h,l\}}\sum_{n \in \{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n}$$
$$\theta = \frac{\ln(I + \tau) - \ln(I)}{\ln(\lambda)}$$

Sales share of high productivity type

$$S = \frac{Sx^h + (\tau - Sx^h - (1 - S)x^l)p^h}{\tau}$$

The expressions related to the labor market clearing condition are derived in Section C.2. This constitutes a system of seven equations in seven unknowns $(x^h, x^l, I, \tau, \frac{Y_t}{w_t}, S)$, which I solve using a root finder.

C.5 Firm markups

Firm markups are defined by $\mu_f = \frac{py_f}{wl_f} = \left(\frac{1}{n}\sum_{k=1}^{n}\mu_{kf}^{-1}\right)^{-1}$. Therefore

$$\ln \mu_f = -\ln\left(\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1}\right).$$

Rewrite the term in brackets (for a high-productivity firm) as

$$\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1} = \frac{1}{n}\sum_{k=1}^{n}e^{-\ln\mu_k} = \frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda} + \sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right), \tag{A-7}$$

where i indexes the product lines where the high productivity firm faces a low productivity second best producer, j the lines where it faces a high productivity second best producer and $n_i + n_j = n$. A two-dimensional linear Taylor expansion around $\ln\lambda = 0$ and $\ln\frac{\varphi^h}{\varphi^l} = 0$ gives

$$\frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda} + \sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right) \approx 1 - \left(\frac{1}{n}\sum_{k=1}^{n}\Delta_k\right)\ln\lambda - \frac{n_i}{n}\ln\left(\frac{\varphi^h}{\varphi^l}\right)$$

such that

$$E\left[\ln\mu_f|\text{firm age} = a_f, \varphi^h\right] \approx E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age} = a_f, \varphi^h\right]\ln\lambda + (1-S)\ln\left(\frac{\varphi^h}{\varphi^l}\right),$$

where I have used the fact that (in expectation) the share of the firm's product lines with a low productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the low productivity type. From Peters (2020), we know that

$$E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age} = a_f, \varphi^h\right]\ln\lambda = \left(1 + I \times E[a_P^h|a_f]\right)\ln\lambda,$$

where $E[a_P^h|a_f]$ denotes the average product age of a high-productivity type firm conditional on firm age a_f and

$$
\begin{aligned}
E[a_P^h|a_f] &= \frac{1}{x^h}\left(\frac{\frac{1}{\tau}\left(1-e^{-\tau a_f}\right)}{\frac{1}{x^h+\tau}\left(1-e^{-(x^h+\tau)a_f}\right)}-1\right)\left(1-\phi^h(a_f)\right)+a_f\phi^h(a_f)\\
\phi^h(a) &= e^{-x^h a}\frac{1}{\gamma^h(a)}\ln\left(\frac{1}{1-\gamma^h(a)}\right)\\
\gamma^h(a) &= \frac{x^h\left(1-e^{-(\tau-x^h)a}\right)}{\tau-x^h e^{-(\tau-x^h)a}}.
\end{aligned}
$$

For a firm of the low-productivity type, the last term in eq. (A-7) reads

$$
\frac{1}{n}\left(\sum_{i=1}^{n_i} e^{-\ln\Delta_i \ln\lambda}+\sum_{j=1}^{n_j} e^{-\ln\frac{\varphi^l}{\varphi^h}-\ln\Delta_j\ln\lambda}\right),
$$

where i indexes the product lines where the low-productivity producer faces a low-productivity second best producer, j the lines where it faces a high-productivity second best producer and $n_i+n_j=n$. Following the same steps as for a high-productivity firm, this time linearizing around $\ln\frac{\varphi^l}{\varphi^h}=0$ (and $\ln\lambda=0$) gives

$$
E\left[\ln\mu_f|\text{firm age}=a_f,\varphi^l\right]\approx\left(1+I\times E[a_P^l|a_f]\right)\ln\lambda+S\ln\left(\frac{\varphi^l}{\varphi^h}\right),
$$

where again I have made use of the fact that (in expectation) the share of the firm's product lines with a high-productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the high-productivity type. $E[a_P^l|a_f]$ is exactly defined as $E[a_P^h|a_f]$ with x^h replaced by x^l in the above expressions.

C.6 Firm size distribution

The mass of high- and low-productivity type firms with $n\geq 2$ products follows the differential equations

$$
\begin{aligned}
\dot{M}_t^h(n) &= (n-1)x_t^h M_t^h(n-1)+(n+1)\tau_t M_t^h(n+1)-n(x_t^h+\tau_t)M_t^h(n)\\
\dot{M}_t^l(n) &= (n-1)x_t^l M_t^l(n-1)+(n+1)\tau_t M_t^l(n+1)-n(x_t^l+\tau_t)M_t^l(n), \qquad \text{(A-8)}
\end{aligned}
$$

whereas the mass of firms with one product evolves according to

$$\dot{M}_t^h(1) = z_t p^h + 2\tau_t M_t^h(2) - (x_t^h + \tau_t) M_t^h(1)$$
$$\dot{M}_t^l(1) = z_t(1-p^h) + 2\tau_t M_t^l(2) - (x_t^l + \tau_t) M_t^l(1). \tag{A-9}$$

The mass of firms with n products increases through firms with $n-1$ products expanding to size n at rate x_t^h or x_t^l per product or through firms with $n+1$ products losing a product at the rate of aggregate creative destruction τ_t. The mass of firms with n products decreases through firms with n products either gaining or losing a product through expansion or creative destruction. The mass of firms with one product additionally increases through firm entry.

Proposition A-1. *The stationary firm size distribution along the balanced growth path is characterized as follows.*

1. *The mass of high and low productivity firms with n products is*

$$M^h(n) = \frac{(x^h)^{n-1} z p^h}{n\tau^n} = \frac{zp^h}{x^h}\frac{1}{n}\left(\frac{x^h}{\tau}\right)^n$$
$$M^l(n) = \frac{(x^l)^{n-1} z(1-p^h)}{n\tau^n} = \frac{z(1-p^h)}{x^l}\frac{1}{n}\left(\frac{x^l}{\tau}\right)^n.$$

2. *The total mass of firms with n products is*

$$M(n) = M^h(n) + M^l(n) = \frac{(x^h)^{n-1} z p^h + (x^l)^{n-1} z(1-p^h)}{n\tau^n}.$$

3. *The mass of firms of each productivity type is*

$$M^h = \sum_{n=1}^{\infty} M^h(n) = \frac{zp^h}{x^h}\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{x^h}{\tau}\right)^n = \frac{zp^h}{x^h}\ln\left(\frac{\tau}{\tau - x^h}\right)$$
$$M^l = \sum_{n=1}^{\infty} M^l(n) = \frac{z(1-p^h)}{x^l}\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{x^l}{\tau}\right)^n = \frac{z(1-p^h)}{x^l}\ln\left(\frac{\tau}{\tau - x^l}\right)$$

4. *The total mass of firms is*

$$M = M^h + M^l.$$

Proof. These results follow from setting the time derivatives in equations (A-8) and (A-9) equal to zero and solving the system of equations. □

For each firm type, the share of firms with n products, $M^h(n)/M^h$ and $M^l(n)/M^l$, follows the PDF of a logarithmic distribution with parameter x^h/τ and x^l/τ as in Lentz and Mortensen (2008). The firm size distribution is highly skewed to the right.

Since there is a continuum of mass one of products and each product is mapped to one firm $\sum_{i=1}^{\infty} M(n) \times n = 1$. Further, the mass of high-productivity type firms producing n products is related to the share of product lines operated by high-type firms, S, as follows

$$S = \sum_{n=1}^{\infty} M^h(n) \times n = \frac{zp^h}{\tau - x^h}.$$

D Computation of transition dynamics

In this section, I lay out the numerical procedure to solve for the transition path. Since time is continuous, I solve a discretized version of the model where the solution converges to the one in continuous time for small enough time intervals. As shown in Appendix C, value functions are additive across product lines. Therefore, I solve the problem of two representative one-product firms: one of the high productivity type and one of the low productivity type.

I normalize the value function by the wage w_t to obtain a stationary problem. The value function for the high-type firm (in discrete time) reads

$$\begin{aligned}
\frac{V_t^h(1,\mu_i,S_t)}{w_t} &= \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^h}\left\{x_t^h \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^h(1,\lambda,S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^h(1,\lambda\frac{\varphi^h}{\varphi^l},S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^h)^\zeta dt\right\} \\
&+ \max_{I_t^h}\left\{I_t^h \exp(-r_{t+dt}dt)\left(\frac{V_{t+dt}^h(1,\mu_i\lambda,S_{t+dt})}{w_{t+dt}} - \frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_I}\mu_i^{-1}(I_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned} \tag{A-10}$$

The value function for the low-type firm reads

$$\begin{aligned}
\frac{V_t^l(1,\mu_i,S_t)}{w_t} &= \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^l}\left\{x_t^l \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^l(1,\lambda\frac{\varphi^l}{\varphi^h},S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^l(1,\lambda,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^l)^\zeta dt\right\} \\
&+ \max_{I_t^l}\left\{I_t^l \exp(-r_{t+dt}dt)\left(\frac{V_{t+dt}^l(1,\mu_i\lambda,S_{t+dt})}{w_{t+dt}} - \frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_I}\mu_i^{-1}(I_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned} \tag{A-11}$$

From this, one obtains the first order conditions for the policy functions. For the optimal expansion R&D rate of the high type firm x_t^h (again suppressing the dependence of the value function on S_t):

$$\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V^h_{t+dt}(1,\lambda)}{w_{t+dt}}+(1-S_{t+dt})\frac{V^h_{t+dt}(1,\lambda\frac{\varphi^h}{\varphi^l})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_x}(x^h_t)^{\zeta-1} \tag{A-12}$$

and for the low type firm x^l_t:

$$\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V^l_{t+dt}(1,\lambda\frac{\varphi^l}{\varphi^h})}{w_{t+dt}}+(1-S_{t+dt})\frac{V^l_{t+dt}(1,\lambda)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_x}(x^l_t)^{\zeta-1}. \tag{A-13}$$

Both are independent of the markup μ_i. For the optimal internal R&D rates of the high type, I^h_t, one obtains

$$\exp(-r_{t+dt}dt)\left(\frac{V^h_{t+dt}(1,\mu_i\lambda)}{w_{t+dt}}-\frac{V^h_{t+dt}(1,\mu_i)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_I}\mu_i^{-1}(I^h_t)^{\zeta-1} \tag{A-14}$$

and similarly for I^l_t

$$\exp(-r_{t+dt}dt)\left(\frac{V^l_{t+dt}(1,\mu_i\lambda)}{w_{t+dt}}-\frac{V^l_{t+dt}(1,\mu_i)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_I}\mu_i^{-1}(I^l_t)^{\zeta-1}. \tag{A-15}$$

Equations (A-10) to (A-15) characterize the firm problem in discrete time. These equations are supplemented by the law of motion for the two dimensional distribution of quality and productivity gaps

$$\nu_{t+dt}\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu_t\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)=dt\left[I_{\mu_i,t}\nu_t\left(\Delta-1,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu_t\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)(I_{\mu_i,t}+\tau_t)\right]\quad\text{for}\quad\Delta\geq 2 \tag{A-16}$$

and for product lines with a unitary quality gap, $\Delta=1$,

$$\nu_{t+dt}\left(1,\frac{\varphi^l}{\varphi^h}\right)-\nu_t\left(1,\frac{\varphi^l}{\varphi^h}\right)=dt\left[(1-S_t)x^l_tS_t+z_t(1-p^h)S_t-\nu_t\left(1,\frac{\varphi^l}{\varphi^h}\right)(I_{\mu_i,t}+\tau_t)\right]$$
$$\nu_{t+dt}\left(1,\frac{\varphi^l}{\varphi^l}\right)-\nu_t\left(1,\frac{\varphi^l}{\varphi^l}\right)=dt\left[(1-S_t)x^l_t(1-S_t)+z_t(1-p^h)(1-S_t)-\nu_t\left(1,\frac{\varphi^l}{\varphi^l}\right)(I_{\mu_i,t}+\tau_t)\right]$$
$$\nu_{t+dt}\left(1,\frac{\varphi^h}{\varphi^h}\right)-\nu_t\left(1,\frac{\varphi^h}{\varphi^h}\right)=dt\left[S_tx^h_tS_t+z_tp^hS_t-\nu_t\left(1,\frac{\varphi^h}{\varphi^h}\right)(I_{\mu_i,t}+\tau_t)\right]$$

$$\nu_{t+dt}\left(1,\frac{\varphi^h}{\varphi^l}\right)-\nu_t\left(1,\frac{\varphi^h}{\varphi^l}\right)=dt\left[S_t x_t^h(1-S_t)+z_t p^h(1-S_t)-\nu_t\left(1,\frac{\varphi^h}{\varphi^l}\right)(I_{\mu_i,t}+\tau_t)\right] \tag{A-17}$$

and a standard Euler equation

$$\frac{C_{t+dt}}{C_t}=\exp(-\rho dt)(1+r_{t+dt}dt). \tag{A-18}$$

Further, the (static) free entry and labor market clearing conditions remain unchanged and are characterized in the main text by equations (8) and (9).

The algorithm to compute the transition path assumes that an initial and ending balanced growth path have been solved for including the (stationary) two-dimensional distribution of quality and productivity gaps. I choose $dt = 0.02$ and set the transition period to 300 years (T), after which I assume the economy has reached its new balanced growth path. I further truncate the two dimensional distribution of quality and productivity gaps along the quality dimension at $\Delta = 30$, implying a maximum quality gap of λ^{30}. No mass reaches this state during the transition such that this assumption is satisfied. I then compute the transition path as follows:

1. Guess a path of wage growth $\frac{w_{d+dt}}{w_t}$ and income to wage ratios Y_t/w_t over the transition (equal to their values in the final balanced growth path)

   (a) Guess a path for S_t over the transition (equal to its value in the final balanced growth path).

      i. Starting backwards in period T, solve for optimal policy functions in $T - dt$ using equations (A-12)-(A-15).[21]

      ii. Solve for τ_{T-dt} that ensures that the free entry condition (8) holds.

      iii. Compute the value function in $T - dt$ using equations (A-10) and (A-11).

      iv. Iterate backwards until the first time period.

[21] I solve for the optimal I_i (at each point in time) over a two-dimensional grid of quality and productivity gaps.

v. Starting from the initial balanced growth path, simulate S_t forward using[22]

$$S_{t+dt} = S_t + dt\left[S_t x_t^h(1-S_t) - (1-S_t)x_t^l S_t + z_t(p^h(1-S_t) - (1-p^h)S_t)\right],$$

where z_t can be substituted out by equation (17).

(b) Update the guess for S_t from step v using bisection and go back to step i. Iterate until the guessed path for S_t converges to the implied one.

2. Starting from the initial balanced growth path, simulate the two dimensional distribution of quality and productivity gaps forward using equations A-16 and A-17.
3. Solve for the implied sequence of $\frac{Y_t}{w_t}$ from the labor market clearing condition.
4. Compute the sequence of quality growth using

$$\frac{Q_{t+dt}}{Q_t} = \exp\left(\left[\int_0^1 I_{\mu_i,t+dt}di + S_{t+dt}x_{t+dt}^h + (1-S_{t+dt})x_{t+dt}^l + z_{t+dt}\right] dt\ln(\lambda)\right).$$

5. Compute the sequence of aggregate productivity growth using

$$\frac{\Phi_{t+dt}}{\Phi_t} = \left(\frac{\varphi^h}{\varphi^l}\right)^{S_{t+dt}-S_t}.$$

6. Using the two dimensional distribution of quality and productivity gaps, compute the sequence of $\mathcal{M}_t$ defined in equation (7).
7. Compute the sequence of production labor L_{Pt} using equation (5).
8. Compute the sequence of aggregate output growth $\frac{Y_{t+dt}}{Y_t}$ using equation (7).
9. With the paths of aggregate output growth and $\frac{Y_t}{w_t}$, obtain the implied path of wage growth $\frac{w_{d+dt}}{w_t}$.
10. Update the guesses for $\frac{w_{d+dt}}{w_t}$ and $\frac{Y_t}{w_t}$ using bisection and go back to step (a). Iterate until the guessed and implied paths converge.

I declare convergence when the difference between the three guessed and implied sequences is less than $\epsilon = 10^{-5}$ in absolute terms. I annualize wage growth before applying the convergence threshold.

[22]One could already simulate the entire two-dimensional distribution of quality and productivity gaps forward here. However, for the inner loop, it is sufficient to iterate on S_t.

E Alternative decomposition

I decompose the contribution of each parameter change to the change in the targeted moments across the initial and new balanced growth path as follows

$$\frac{M_j\left(\boldsymbol{\theta}_{-i}^{new},\theta_i^{new}\right)-M_j\left(\boldsymbol{\theta}_{-i}^{new},\theta_i^{initial}\right)}{M_j\left(\boldsymbol{\theta}_{-i}^{new},\theta_i^{new}\right)-M_j\left(\boldsymbol{\theta}_{-i}^{initial},\theta_i^{initial}\right)}\times 100, \tag{A-19}$$

where *initial* and *new* index the initial and new balanced growth path.[23] The difference to the decomposition in the main text is that eq. (A-19) quantifies the contribution of each parameter change starting from the new balanced growth path. Table A-1 shows the contribution of each parameter change (rows) to the five moments of the estimation (columns). The notation follows Table 3.

Table A-1: Alternative decomposition

	VA	wL	σVA	$\frac{z}{M}$	g
Parameter Δ					
Expansion R&D efficiency $\psi_x \uparrow$	240.8	228.2	133.7	219.3	-5.8
Internal R&D efficiency $\psi_I \uparrow$	-20.3	-20.2	-100.0	-123.1	-23.2
Productivity gap $\varphi^h/\varphi^\ell \uparrow$	29.4	26.8	61.5	94.4	8.4
Entry R&D efficiency $\psi_z \uparrow$	-627.3	-580.0	-2276.4	-2711.9	-170.3
Step size of quality improvements $\lambda \downarrow$	71.1	68.5	116.9	422.3	256.5

Notes: the table shows the contribution of each parameter change (rows) to the changes in the five moments (columns) across the intial and new balanced growth path (BGP) in percent. The contribution of each parameter change is computed using eq. (A-19). Moments and parameter values of the initial and new BGP are shown in Table 4. VA denotes the difference in average log value added between firms aged eight and entrants, wL the difference in the average log wage bill between firms aged eight and entrants, σVA the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate.

Table A-1 confirms the results of the decomposition in the main text. The increase in firm size relative to entry is mainly driven by the increase in the expansion R&D efficiency and the fall in the step size of quality improvements. The decomposition in Table A-1 attributes a somewhat larger role to the fall in the step size of quality improvements in explaining the increase in firm size relative to entry than the one starting from the initial balanced growth path in the main text. The relationship between firm entry and entry R&D costs is steeper for low levels of entry, which explains the increased contribution of the entry R&D costs in

[23] Setting ψ_I or ψ_z to their values in the initial balanced growth path with all other parameters set at their new balanced growth path values leads to zero firm entry. Without entry, the firm-size distribution is non-stationary, i.e., no stationary equilibrium exists. Hence, for ψ_I or ψ_z, I compute the contribution by multiplying the contribution of a 0.1% change of the total change in the respective parameter by a factor of 1000.

Table A-1 relative to the previous decomposition. However, the role of each parameter in explaining the targeted moment changes is the same as in the previous decomposition.